Wireless sensor network UNIT-I

Sixth Sem

ECE-342T

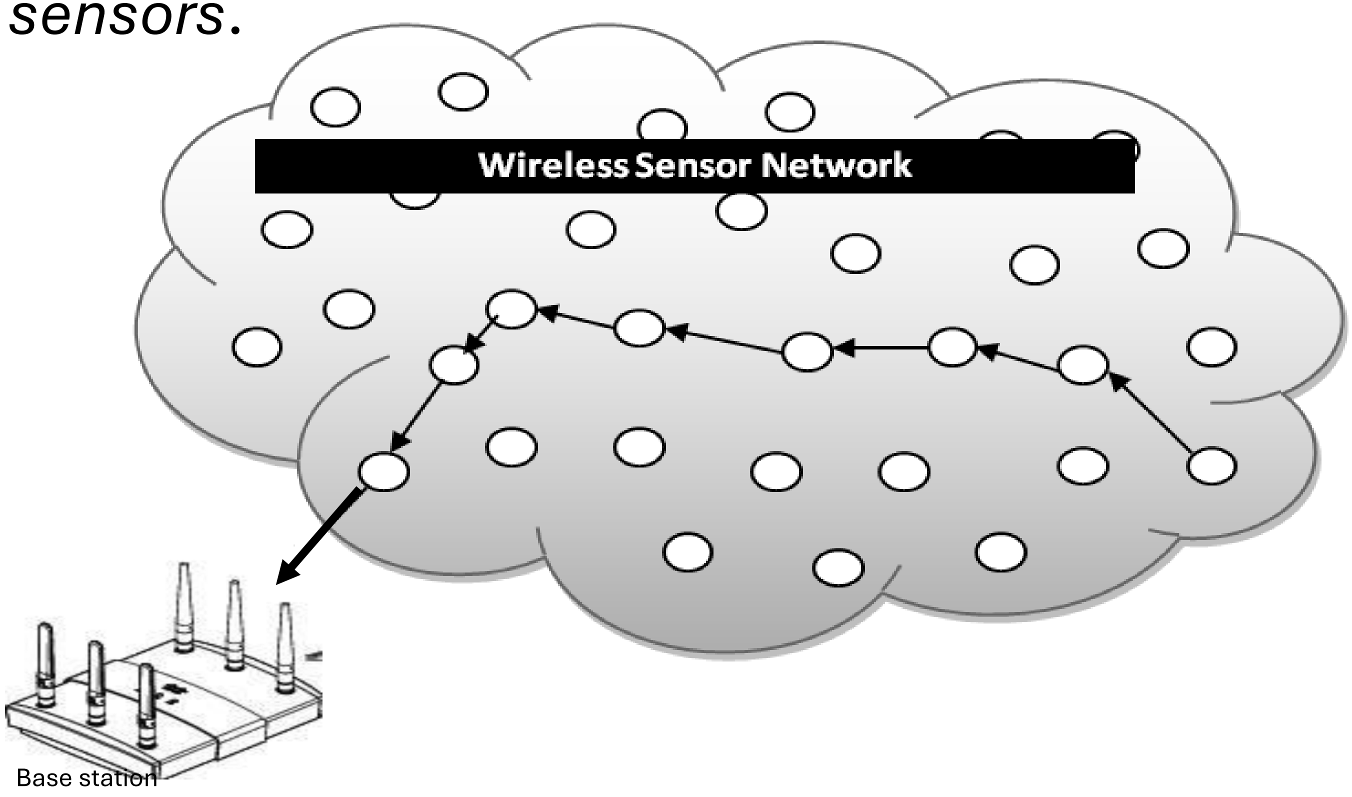
INTRODUCTION TO WIRELESS SENSOR NODE

Wireless Sensor Node

- sensor
  - A transducer
  - converts physical phenomenon e.g. heat, light, motion, vibration, and sound into electrical signals
- sensor node
  - basic unit in sensor network
  - contains on-board sensors, processor, memory, transceiver, and power supply
- sensor network
  - consists of a large number of sensor nodes
  - nodes deployed either inside or very close to the sensed phenomenon

Wireless Sensor Networks (WSNs)

A sensor network is a wireless network that consists of thousands of very small nodes called *sensors*.



Architecture of wireless sensor networks

Introduction

- sensor devices
    - low-cost
    - low-power
    - Multifunctional
- A sensor network that can provide access to information anytime, anywhere by collecting, processing, analyzing and disseminating data.

Introduction （cont’）

- Sensor networks promise to revolutionize sensing in a wide range of application domains.
  - reliability
  - accuracy
  - flexibility
  - Cost-effectiveness
  - ease of deployment

Introduction （cont’）

- Sensor networks enable：
  - information gathering
  - information processing
  - reliable monitoring

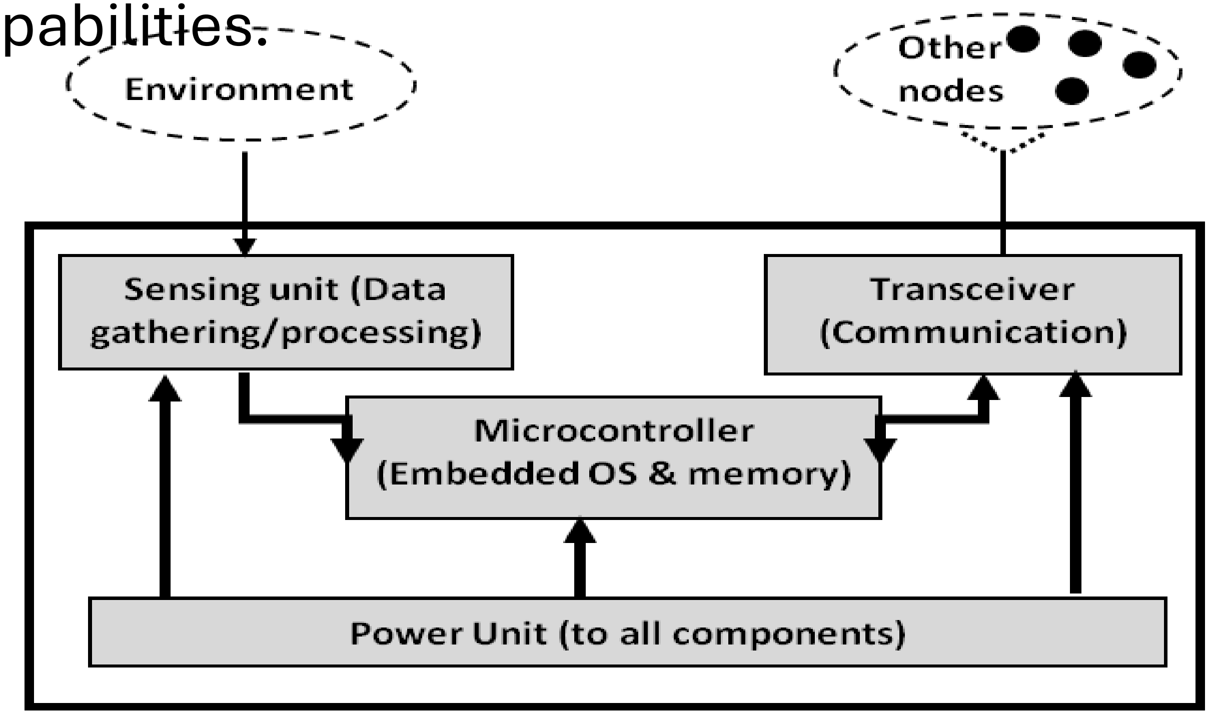
  of a variety of environments for both civil and military applications.

Introduction （cont’）

- The architecture of the sensor node’s hardware consists of five components:
  - sensing hardware
  - Processor
  - memory
  - power supply
  - transceiver
- These devices are easily deployed
  - no infrastructure and human control are needed

Wireless Sensor Networks (cont.)

WSN **Sensors** are equipped with sensing, limited computation, and wireless communication capabilities.



Typical hardware components of a sensor node in wireless sensor networks

Introduction （cont’）

- Each sensor node has
  - wireless communication capability
  - sufficient intelligence for signal processing and for disseminating the data
- Communication in sensor networks is not typically end to end.
- Wireless network
- Energy is typically more limited in sensor networks. — difficulty in recharging

Introduction （cont’）

- Bluetooth devices are unsuitable for sensor network applications
  - because of their energy requirements
  - and expected higher costs than sensor nodes
- a denser infrastructure would lead to a more effective sensor network.
  - It can provide higher accuracy
  - and has a larger aggregate amount of energy available

Introduction （cont’）

- if not properly managed, a denser network can intelligence for signal processing and also lead to a larger number of collisions and potentially to **congestion** in the network
  - increase latency
  - reduce energy efficiency

Examples of possible applications

- Sensors are deployed to analyze remote locations
  - the motion of a tornado(cyclone)
  - fire detection in a forest
- Sensors are attached to taxi cabs in a large metropolitan area to study the traffic conditions and plan routes effectively.
- Wireless parking lot sensor networks that determine which spots are occupied and which spots are free.
- Wireless surveillance sensor networks for providing security in a shopping mall, parking garage or at some other facility.
- Military sensor networks to detect, locate or track enemy movements.
- Sensor networks can increase alertness to potential terrorist threats.

Introduction

- Wireless Sensor Networks are networks that consists of sensors which are distributed in an ad hoc manner.
- These sensors work with each other to sense some physical phenomenon and then the information gathered is processed to get relevant results.
- Wireless sensor networks consists of protocols and algorithms with self-organizing capabilities.

WSNs Applications

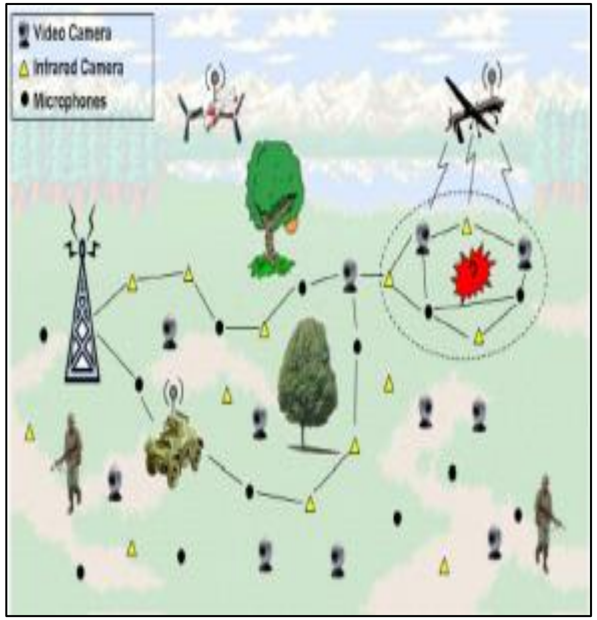
- WSNs have many advantages over traditional networking techniques.

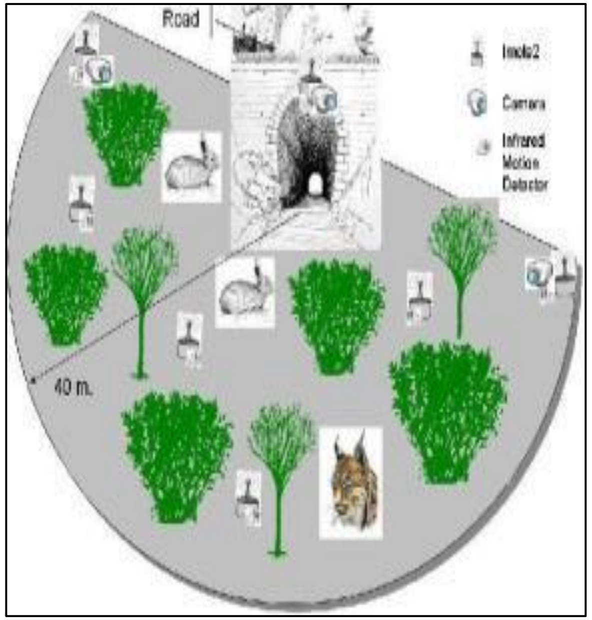
- They have an increasing number of applications, such as infrastructure protection and security, surveillance, health-care, environment monitoring, food safety, intelligent transportation, and smart energy.

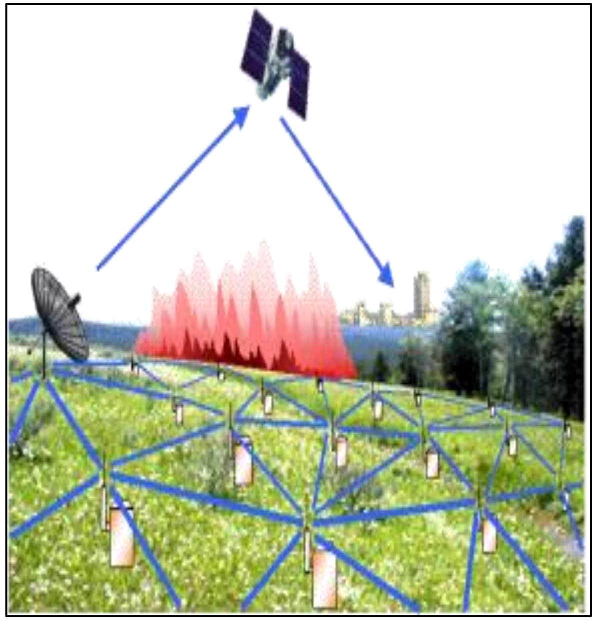
Comparison with ad hoc networks

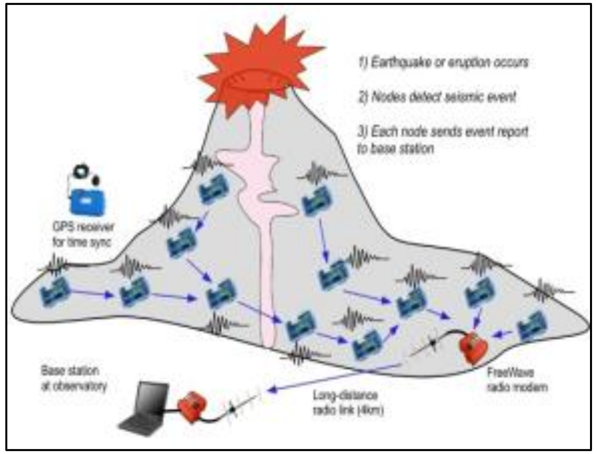
- Wireless sensor networks mainly use **broadcast** communication while ad hoc networks use **point-to-point** communication.
- Unlike ad hoc networks wireless sensor networks are **limited by sensors** limited power, energy and computational capability.
- Sensor nodes may **not have global ID** because of the large amount of overhead and large number of sensors.

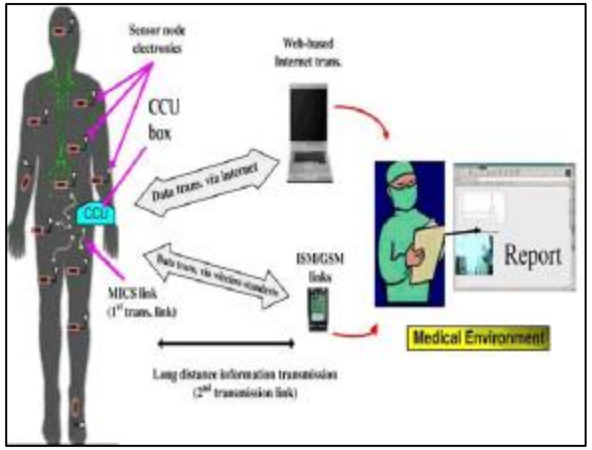
WSNs Applications

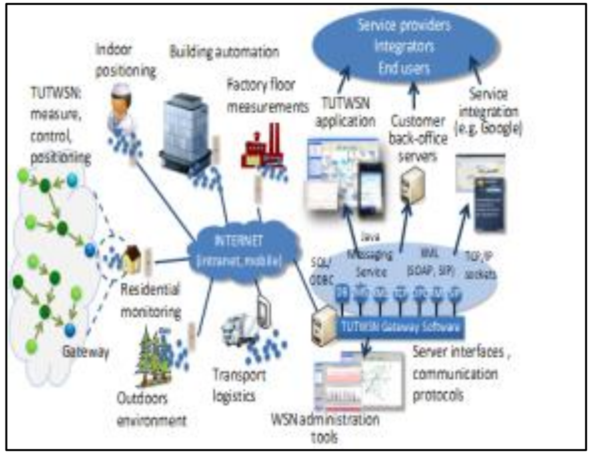












WSNs Applications

Applications of Wireless Sensor networks

The applications can be divided in three categories:

- Monitoring of objects.
- Monitoring of an area.
- Monitoring of both area and objects.

Monitoring Objects

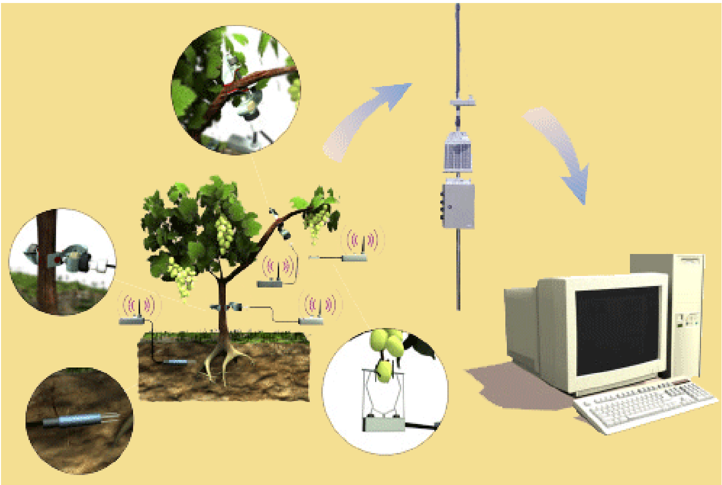
- Structural Monitoring
- Eco-physiology
- Condition-based Maintenance
- Medical Diagnostics
- Urban terrain mapping

Monitoring Area

- Environmental Monitoring
- Precision Agriculture
- Indoor Climate Control
- Military Surveillance
- Intelligent Alarms

Precision Agriculture

- Precision agriculture aims at making cultural operations more efficient, while reducing environmental impact.
- The information collected from sensors is used to evaluate optimum sowing density, estimate fertilizers and other inputs needs, and to more accurately predict crop yields.



Monitoring Interactions between Objects and Space

- Wildlife Habitats
- Disaster Management
- Emergency Response
- Ubiquitous Computing
- Asset Tracking
- Health Care
- Manufacturing Process Flows

Characteristics of Wireless Sensor Networks

- Wireless Sensor Networks mainly consists of **sensors. Sensors** are -
  - low power
  - limited memory
  - energy constrained due to their small size.

- Wireless networks can also be deployed in **extreme environmental** conditions and may be prone to enemy attacks.

- Although deployed in an ad hoc manner they need to be **self organized** and **self healing** and can face constant reconfiguration.

Design Challenges for WSN

Heterogeneity

The devices deployed maybe of various types and need to collaborate with each other.

Distributed Processing

The algorithms need to be centralized as the processing is carried out on different nodes.

Low Bandwidth Communication

The data should be transferred efficiently between sensors

Continued..

Large Scale Coordination

The sensors need to coordinate with each other to produce required results.

Utilization of Sensors

The sensors should be utilized in a ways that produce the maximum performance and use less energy.

Real Time Computation

The computation should be done quickly as new data is always being generated.

Operational Challenges of Wireless Sensor Networks

- Energy Efficiency
- Limited storage and computation
- Low bandwidth and high error rates
- Errors are common
  - Wireless communication
  - Noisy measurements
  - Node failure are expected
- Scalability to a large number of sensor nodes
- Survivability in harsh environments
- Experiments are time- and space-intensive

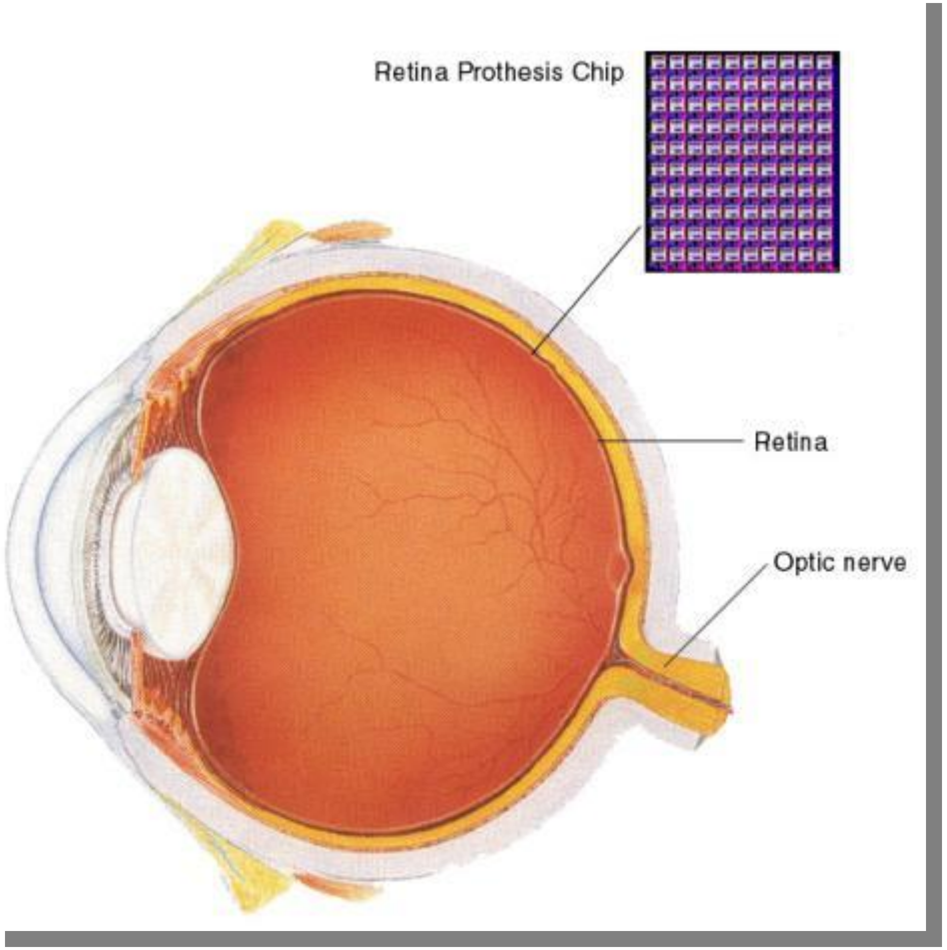
Future of WSN

Smart Home / Smart Office

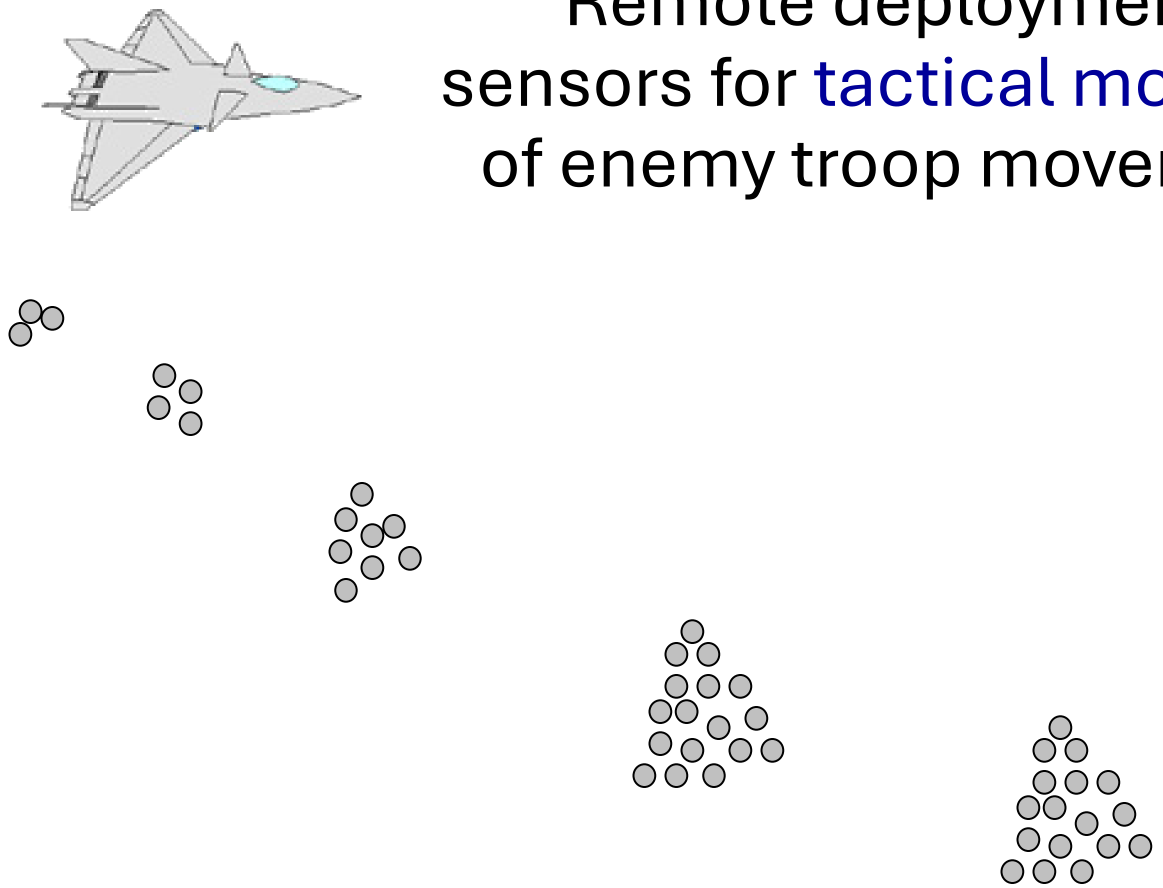
- Sensors controlling electrical devices in the house.
- Better lighting and heating in office buildings.
- The Pentagon building has used sensors extensively.

Biomedical / Medical

- Health Monitors
  - Glucose
  - Heart rate
  - Cancer detection
- Chronic Diseases
  - Artificial retina
  - Cochlear implants
- Hospital Sensors
  - Monitor vital signs
  - Record anomalies



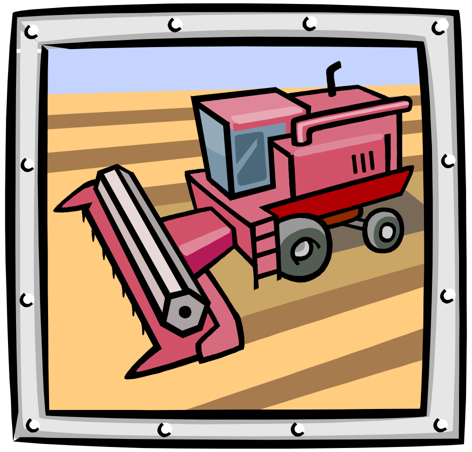
Military



Remote deployment of sensors for tactical monitoring of enemy troop movements.

Industrial & Commercial

- Numerous industrial and commercial applications:
  - Agricultural Crop Conditions
  - Inventory Tracking
  - In-Process Parts Tracking
  - Automated Problem Reporting
  - Theft Deterrent and Customer Tracing
  - Plant Equipment Maintenance Monitoring



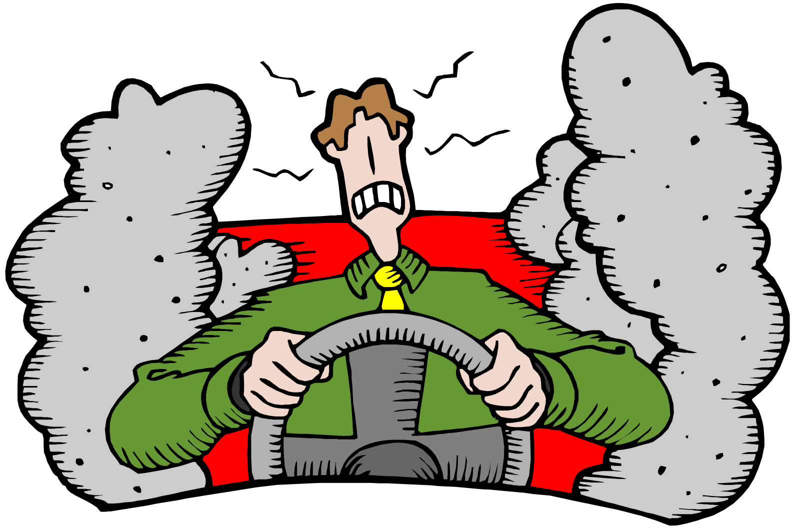
Traffic Management & Monitoring

Future cars could use wireless sensors to:

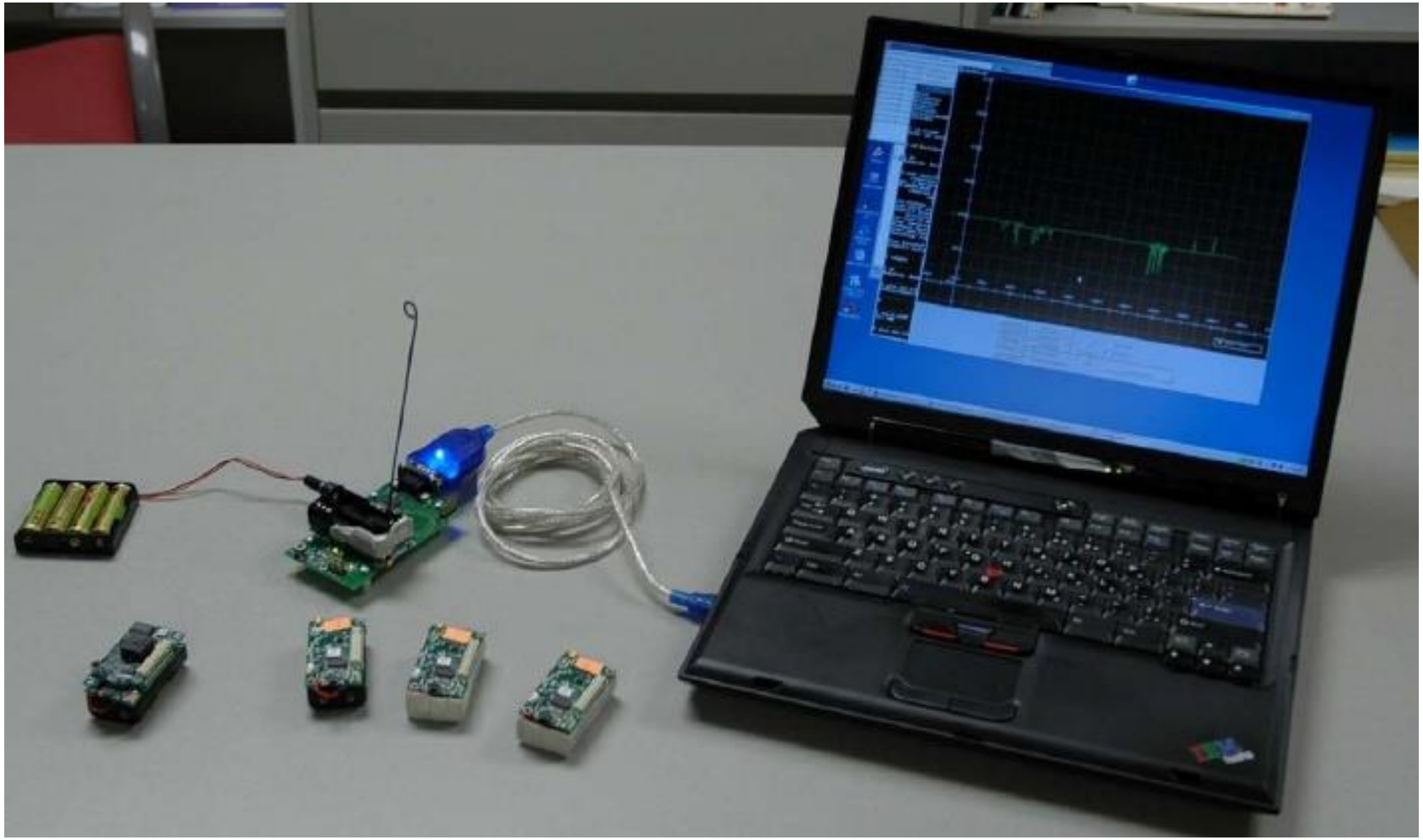
- Handle Accidents
- Handle Thefts

Sensors embedded in the roads to:

- Monitor traffic flows
- Provide real-time route updates



Hardware Setup Overview



Sensor Network Algorithms

- Directed Diffusion – Data centric routing
- Sensor Network Query Processing
- Distributed Data Aggregation
- Localization in sensor networks
- Multi-object tracking/Pursuer Evader
- Security

Wireless Sensor network
VI Sem

MANET(Mobile Ad-hoc Network)

Wireless networking in general:

Wireless networks typically work by one of the configuration network topology either Ad-Hoc or Infrastructure network.

Topology on Network consists of

1. Ad-Hoc Topology

2. Infrastructure Topology

Ad-Hoc Topology

Ad hoc wireless network is a collection of nodes (or router) mobile wireless that dynamically existence without the use of existing network infrastructure or centralized administration.

Ad Hoc wireless network can also be regarded as more decentralized wireless network.

Ad-hoc network is a form of wireless communication networks are the simplest.

Ad-Hoc Topology

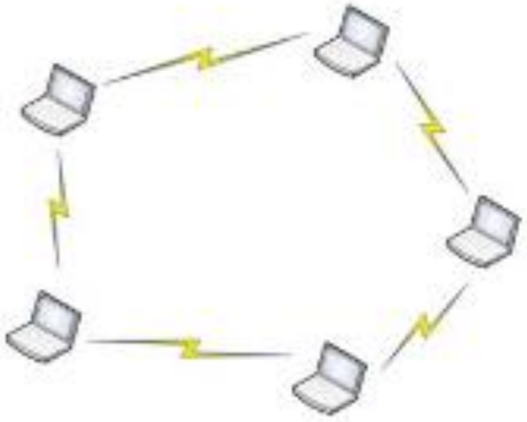
In the Ad Hoc network, the router can freely perform network organization that resulting topology will change quickly and are difficult to predict. With this feature, the Ad-Hoc network experienced several challenges, among others,

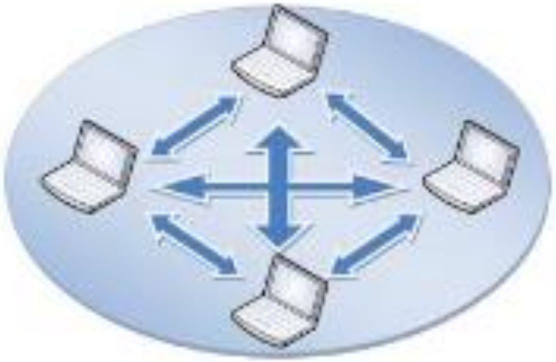
- Multihop
- Mobility
- The combination of large networks with a range of different tools
- Bandwidth
- Limitations of battery consumption

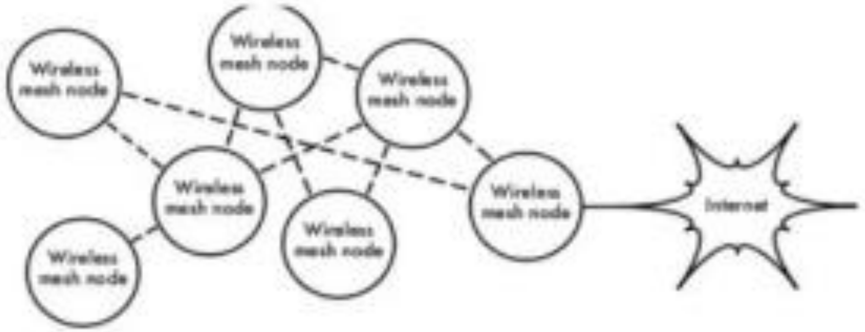
Ad-Hoc Topology

Ad Hoc network also requires a routing protocol because each node requires data exchange.

In contrast to the infrastructure network, ad-hoc networks do not require a wireless LAN to connect each computer and network topology is a mesh network formed.







Ad-Hoc Topology

Here are some of the advantages of a wireless ad-hoc network:

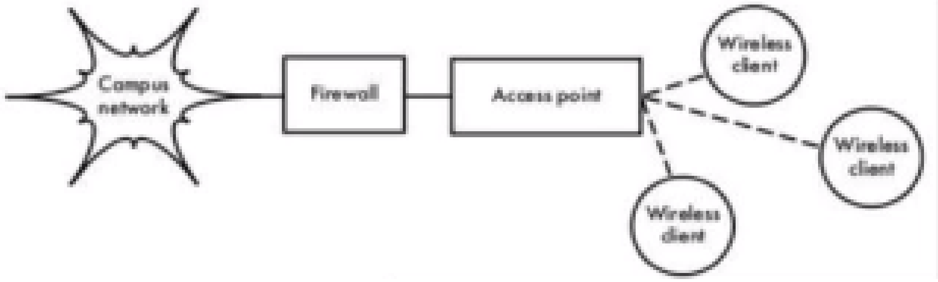
- Ad-Hoc wireless network is very simple in its setup. Plug the wireless adapter into the laptop / computer, configure the software, and you also are able to do communication between laptop
- Ad-Hoc network is cheap because you do not need a wireless access point.
- Ad-Hoc network is fast. Throughput rate between the adapter twice faster than you are using a wireless access point in the infrastructure topology.

Infrastructure topology

* Concept of infrastructure network where it is necessary to establish a wireless LAN network as a center.
* Wireless LANs have the SSID (Service Set Identifier) as the name of the wireless network, with wireless LAN SSID then it can be recognized.
* At the time several computers connected to the same SSID, then formed a network infrastructure.
* It appears that some computers connected by a wireless LAN, here network topology formed is a star topology.

Infrastructure topology

- With an infrastructure network topology allows you to:
  1. Connected to the wired LAN. A wireless access point lets you extend your LAN network with the ability to connect wirelessly.
  2. Computers on a wired network and a computer with a wireless connection can communicate with each other. This had been the main strength of the infrastructure wireless topology.
  3. Extending the range of your wireless. By putting a wireless access point between the two wireless adapters extend the range to be doubled.



MOBILE AD HOC NETWORK

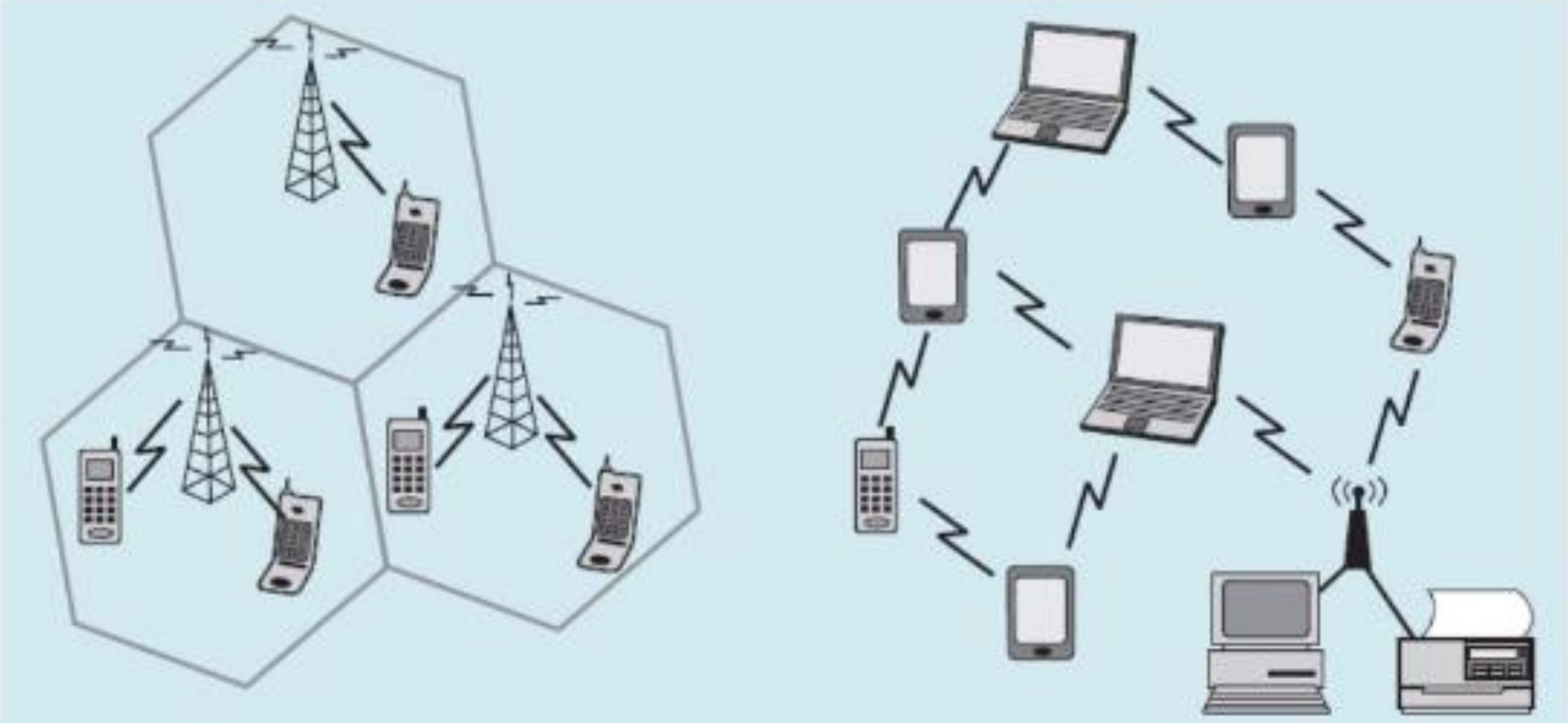
* Mobile Ad Hoc Network (MANET) indicates a wireless network of mobile nodes that have no fixed routers.
* The nodes in this network also serve as routers that are responsible for finding and dealing with the route to every node in the network.
* Some characteristics of MANET are:
  * dynamic network configuration,
  * limited bandwidth
  * power constraints for each operation,
  * low overheads,

  and an adaptive system able to handle packet loss.
* The MANET network layer has two parts, namely the network layer and the transport layer.
* In the network layer of MANET is the IP (Internet protocol) and the ad hoc routing layer uses the AODV protocol (ad hoc on demand distance vector)

MOBILE AD HOC NETWORK

* A mobile ad hoc network is an autonomous collection of mobile devices (laptops, smart phones, sensors, etc.) that communicate with each other over wireless links and cooperate in a distributed manner in order to provide the necessary network functionality in the absence of a fixed infrastructure.
* This type of network, operating as a stand-alone network or with one or multiple points of attachment to cellular networks or the Internet, paves the way for numerous new and exciting applications.
* Application scenarios include, but are not limited to: emergency and rescue operations, conference or campus settings, car networks, personal networking, etc.

MOBILE AD HOC NETWORK



(a) Cellular network

(b) Mobile ad hoc network

Characteristics Interface of MANET

Dynamic Topology: Node in MANET has a dynamic nature, which can be moved anywhere.
Autonomy: Each node in MANET role as **end-user** as well as a router which calculates its own route-path which will subsequently be selected.
Bandwidth limitations: Link on wireless networks tend to have low capacity when compared to a wired network.
Limitations of energy: All nodes in MANET are mobile, so it is ascertained that node uses battery power to operate. So it is necessary to optimize the design of energy.
Limitations Security: Wireless networks tend to be more vulnerable to security than wired networks.

Mobile Ad Hoc Network Enabling Tech.

Technology	Theoretical bit rate	Frequency	Range	Power consumption
IEEE 802.11b	1, 2, 5.5 and 11 Mbit/s	2.4 GHz	25–100 m (indoor) 100–500 m (outdoor)	~30 mW
IEEE 802.11g	Up to 54 Mbit/s	2.4 GHz	25–50 m (indoor)	~79 mW
IEEE 802.11a	6, 9, 12, 24, 36, 49 and 54 Mbit/s	5 GHz	10–40 m (indoor)	40 mW, 250 mW or 1 W
Bluetooth (IEEE 802.15.1)	1 Mbit/s (v1.1)	2.4 GHz	10 m (up to 100 m)	1 mW (up to 100 mW)
UWB (IEEE 802.15.3)	110 – 480 Mbit/s	Mostly 3 – 10 GHz	~10 m	100 mW, 250 mW
IEEE 802.15.4 (for example, Zigbee)	20, 40 or 250 kbit/s	868 MHz, 915 MHz or 2.4 GHz	10–100 m	1 mW
HiperLAN2	Up to 54 Mbit/s	5 GHz	30–150 m	200 mW or 1 W
IrDA	Up to 4 Mbit/s	Infrared (850 nm)	~10 m (line of sight)	Distance based
HomeRF	1 Mbit/s (v1.0) 10 Mbit/s (v2.0)	2.4 GHz	~50 m	100 mW
IEEE 802.16 IEEE 802.16a IEEE 802.16e (Broadband Wireless)	32 – 134 Mbit/s up to 75 Mbit/s up to 15 Mbit/s	10–66 GHz < 11 GHz < 6 GHz	2–5 km 7–10 km (max 50 km) 2–5 km	Complex power control

Ad-Hoc network is divided into 7 Ad Hoc network types are as follows:

WANET (*Wireless Ad Hoc Network*)

MANET (*Mobile Ad Hoc Network*)

VANET (*Vehicular Ad Hoc Network*)

SPANs (*Smart Phone Ad Hoc Network*)

iMANETs (*Internet Based Mobile Ad Hoc Network*)

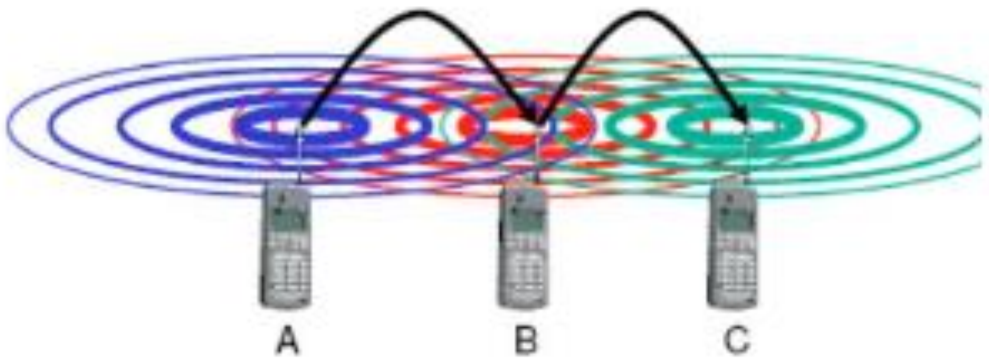
Military / Tactical MANETs

SPAN (*Self Powered Ad Hoc Network*).

MANET (*Mobile Ad Hoc Network*)

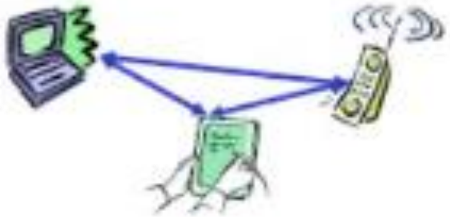
Mobile ad hoc networks

* Standard Mobile IP needs an infrastructure
  - Home Agent/Foreign Agent in the fixed network
  - DNS, routing etc. are not designed for mobility
* Sometimes there is no infrastructure!
  - remote areas, ad-hoc meetings, disaster areas
  - cost can also be an argument against an infrastructure!
* Main topic: routing
  - no default router available
  - every node should be able to forward



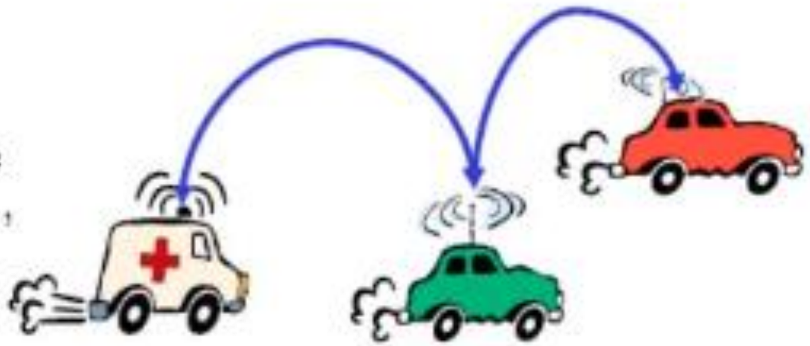
Solution: Wireless ad-hoc networks

Network without infrastructure
Use components of participants for networking

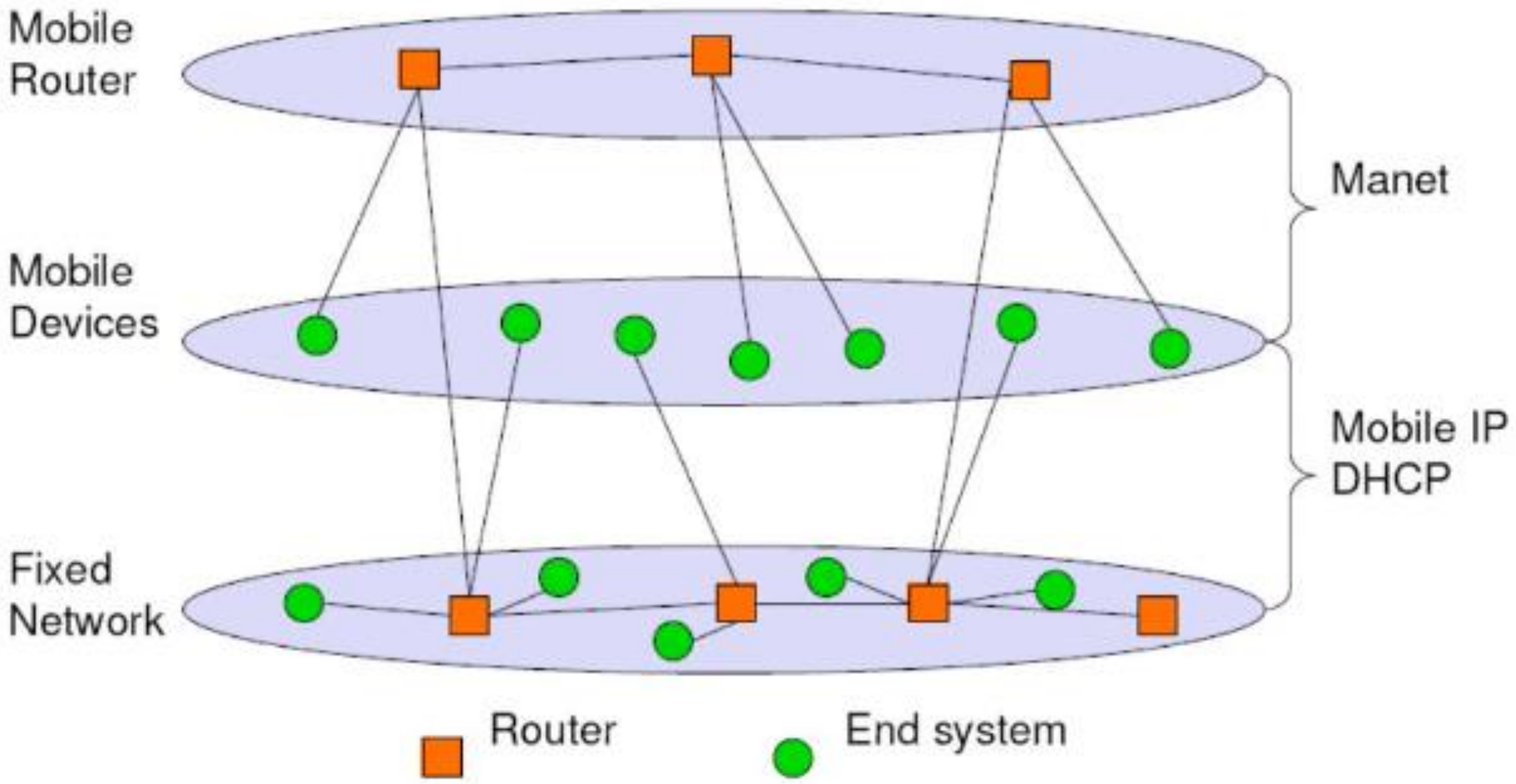


Examples
Single-hop: All partners max. one hop apart
Bluetooth piconet, PDAs in a room,
gaming devices…

Multi-hop (Mesh): Cover larger distances,
circumvent obstacles
Bluetooth scatternet, TETRA police network
car-to-car networks…



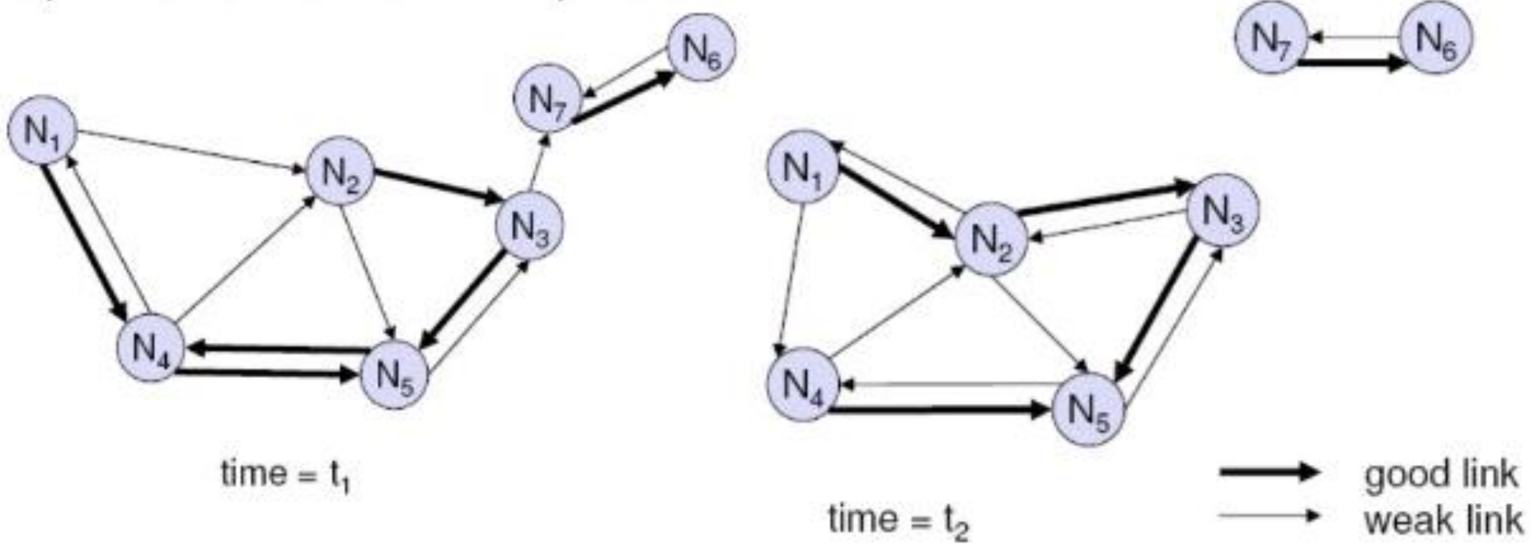
MANET: Mobile Ad-hoc Networking



Problem No. 1: Routing

Highly dynamic network topology

* Device mobility plus varying channel quality
* Separation and merging of networks possible
* Asymmetric connections possible



Problems of traditional routing algorithms

Dynamic of the topology

- frequent changes of connections, connection quality, participants

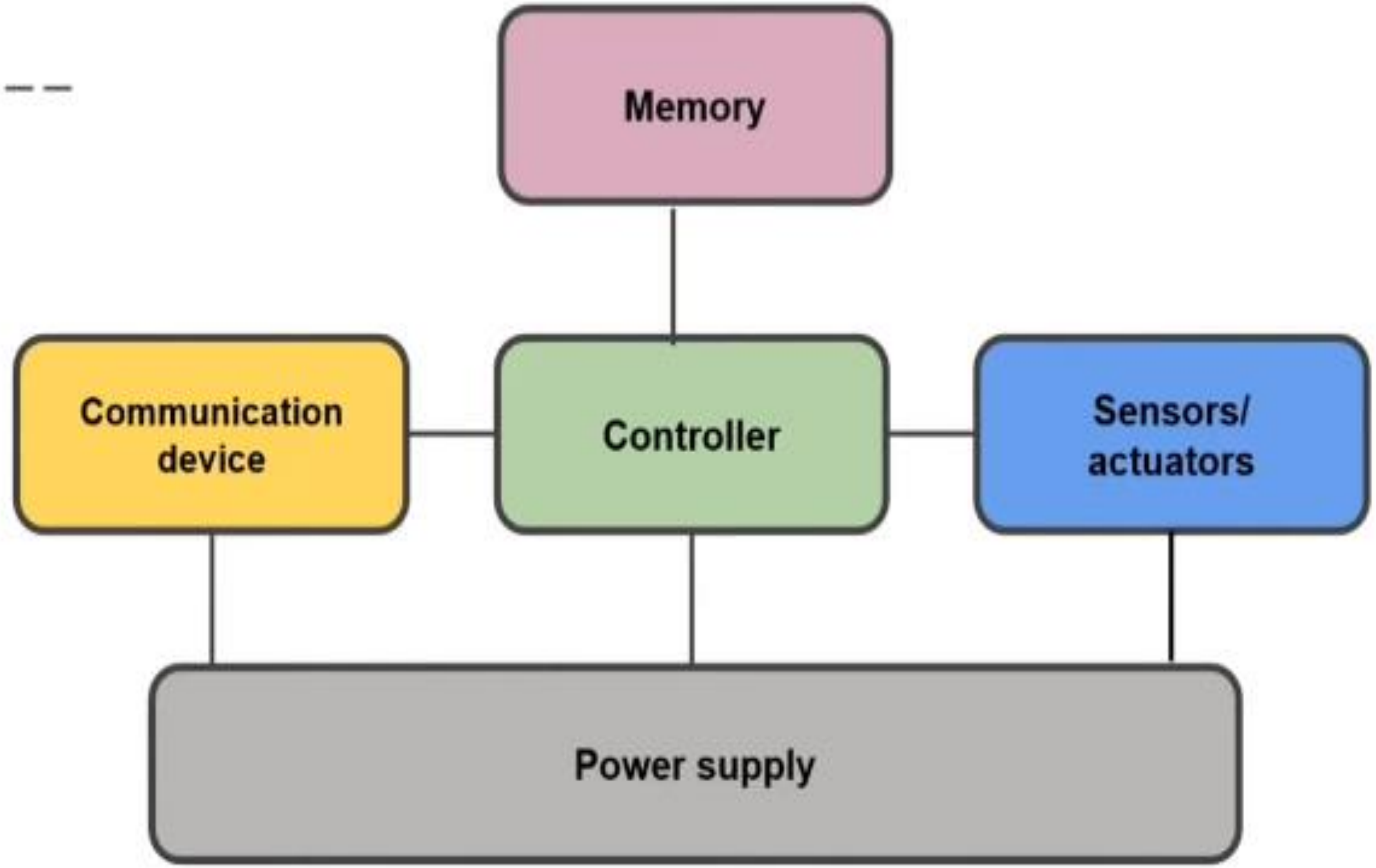
Limited performance of mobile systems

- periodic updates of routing tables need energy without contributing to the transmission of user data, sleep modes difficult to realize
- limited bandwidth of the system is reduced even more due to the exchange of routing information
- links can be asymmetric, i.e., they can have a direction dependent transmission quality

WIRELESS SENSOR NETWORK

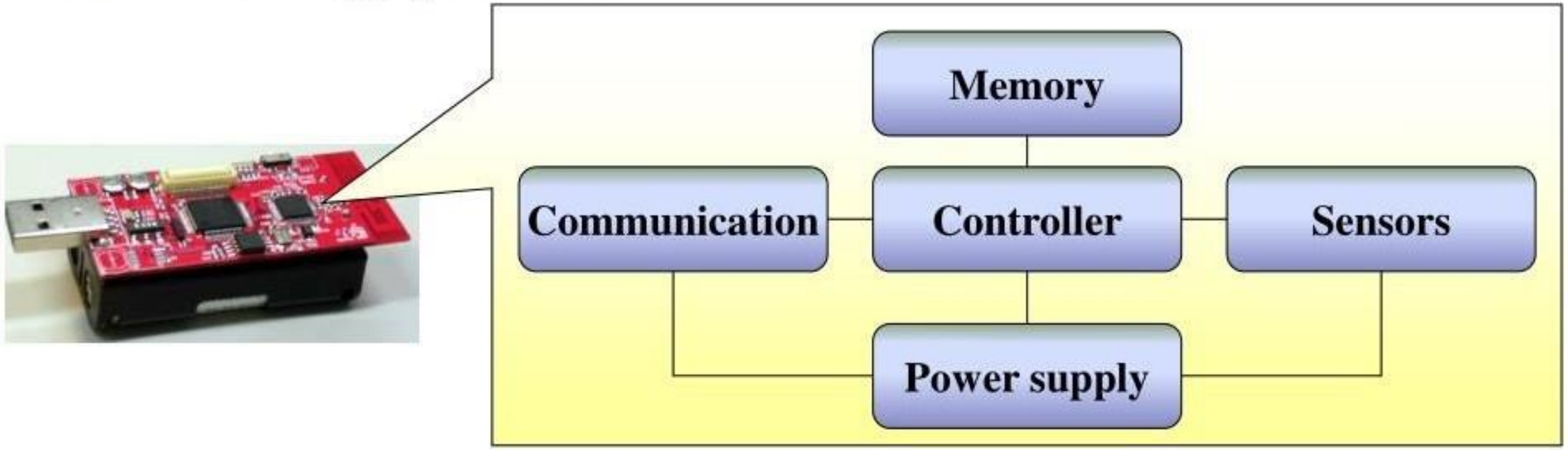
Single Node Architecture

Single Node Architecture



Main Architecture of Sensor Node

- The main architecture of sensor node includes following components:
  - Controller module
  - Memory module
  - Communication module
  - Sensing modules
  - Power supply module



Controller

(a) Microcontroller

- Texas Instruments
- Atmel ATMega
- Intel Strong ARM

(b) PDSPs

- Broadband wireless communication

(C) FPGAs

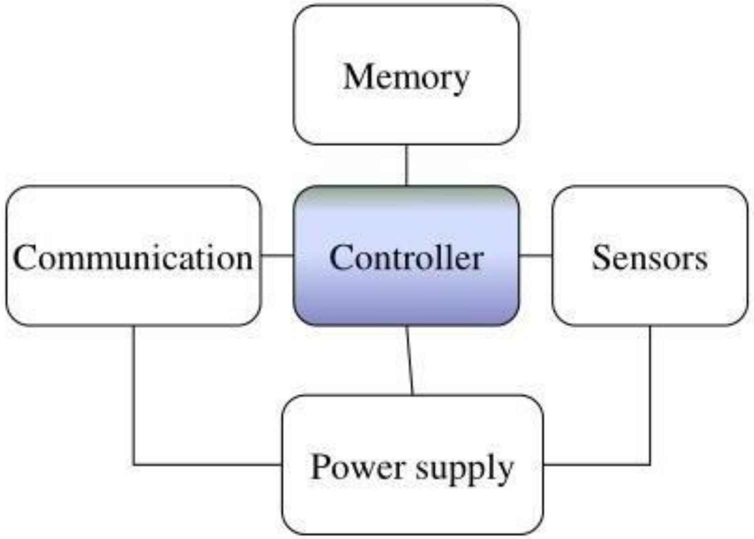
- Good for testing
- Provides high flexibility

(D) ASICs- Custom Designed

- Only when peak performance is needed
- No flexibility

Main Components of a Sensor Node: Controller Module

- Main options:
  - MCUs (Microcontrollers)
    - The processor for general purposes
    - Optimized for embedded applications
    - Low energy consumption
  - DSPs (Digital Signal Processors)
    - Optimized for signal processing
    - Low cost
    - High processing speed
    - Not suitable for sensor node
  - FPGAs (Field Programmable Gate Arrays)
    - Suitable for product development and testing
    - Cost higher than DSPs
    - High energy consumption
    - Processing speed lower than ASICs
  - ASICs (Application-Specific Integrated Circuits)
    - Only when peak performance is needed
    - For special purpose
    - Not flexiable



Communication Device

- _To exchange data between individual nodes.
- Transmission medium
    - Optical Communication
    - Ultrasound
    - Radio frequency
        - Long range
        - High data rates
        - Acceptable error rate
        - Reasonable energy expenditure
        - Does not require line of sight(LOS)

Main Components of a Sensor Node: Communication Module

- Radio performance
  - Modulation
    - ASK, FSK, PSK, QPSK…
  - Noise figure: SNR
  - Gain: the ratio of the output signal power to the input power signal
  - Carrier sensing and RSSI characteristics
  - Frequency stability (Ex: towards temperature changes)
  - Voltage range

Main Components of a Sensor Node: Communication module

- Transceivers typically has several different **states/modes** :
  - **Transmit** mode
    - Transmitting data
  - **Receive** mode
    - Receiving data
  - **Idle** mode
    - Ready to receive, but not doing so
    - Some functions in hardware can be switched off
    - Reducing energy consumption a little
  - **Sleep** mode
    - Significant parts of the transceiver are switched off
    - Not able to immediately receive something
    - Recovery time and startup energy in sleep state can be significant

Sensors & Actuators

Sensors

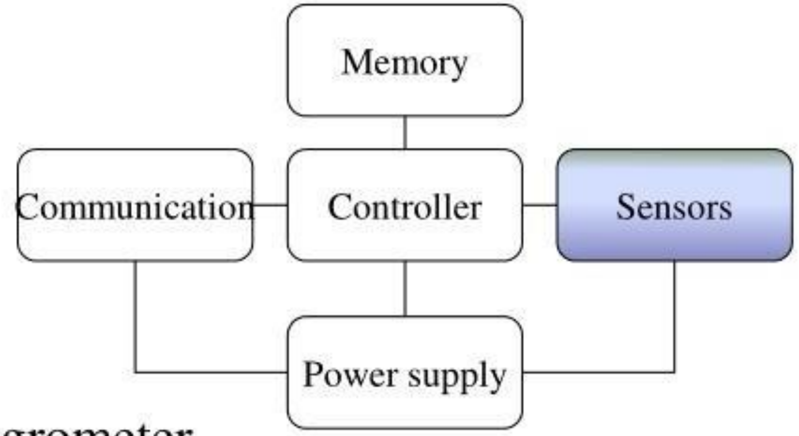
- Actual interface to the physical world
- Observe or control physical parameters of the environment
- Sensors can be roughly categorized into three categories as
    - Passive omnidirectional sensors
    - Passive narrow-beam sensors
    - Active sensors

Actuators

- Converts electrical signals into physical phenomenon.

Main Components of a Sensor Node: Sensing Module

- Sensor's main categories [1]
  - Passive vs. Active
  - Directional vs. Omidirectional
- Some sensor examples
  - Passive & Omnidirectional
    - light, thermometer, microphones, hygrometer, …
  - Passive & Directional
    - electronic compass, gyroscope, …
  - Passive & Narrow-beam
    - CCD Camera, triple axis accelerometer, infar sensor …
  - Active sensors
    - Radar, Ultrasonic, …



Main Components of a Sensor Node: Sensing Module

- Example of sensors are integrated with Senor Node

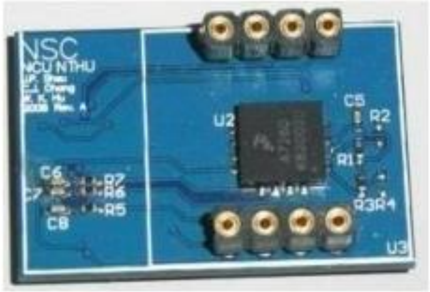
Infar sensor

Ultrasonic

Gyroscope

Electronic compass

Pressure Sensor

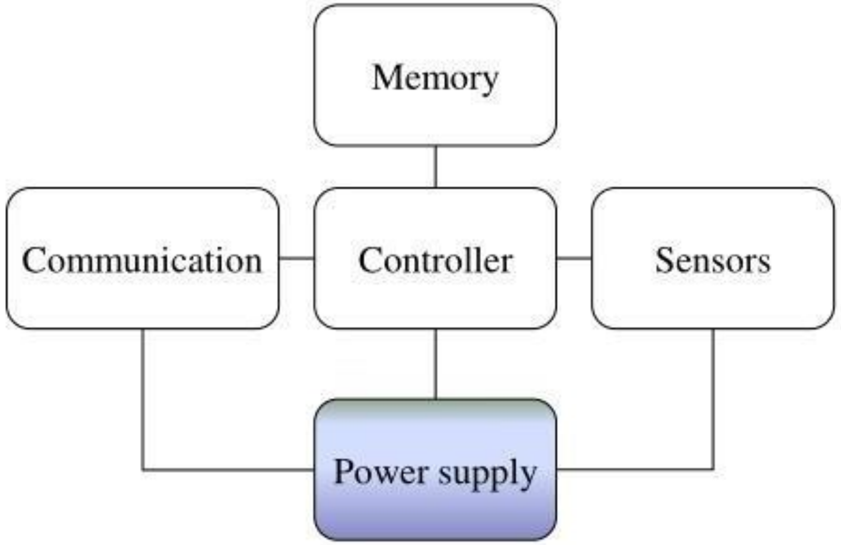


Triple axis accelerometer

Temperature and
Humidity Sensor

Main Components of a Sensor Node: Power supply module

- Power supply module
  - provides as much energy as possible
  - includes following requirements
    - Longevity (long shelf live)
    - Low self-discharge
    - Voltage stability
    - Smallest cost
    - High capacity/volume
    - Efficient recharging at low current
    - Shorter recharge time
- Options of power supply module
  - Primary batteries
    - not rechargeable
  - Secondary batteries
    - rechargeable
    - In WSN, recharging may or may not be an option



Memory

- To store programs and intermediate data
  - Random Access Memory (RAM)
  - Read-Only Memory (ROM)
  - Electrically Erasable Programmable Read-Only Memory (EEPROM)
  - Flash memory.

Introduction of Sensor Hardware Platform

Introduction of Octopus X Hardware Platform

- Octopus X includes three models
  - Octopus X-A
    - CC2431 + Inverted F Antenna
  - Octopus X-B
    - CC2431 + SMA Type Antenna
  - Octopus X-C
    - CC2431 + Inverted F and SMA Type Antenna + USB interface

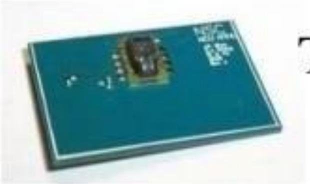
Octopus X-A

Octopus X-B

Octopus X-C

- Peripherals of Octopus X
  - Octopus X-USB dongle
  - Octopus X-Sensor board
    - Temperature sensor
    - Gyroscope
    - Three axis accelerometer
    - Electronic Compass

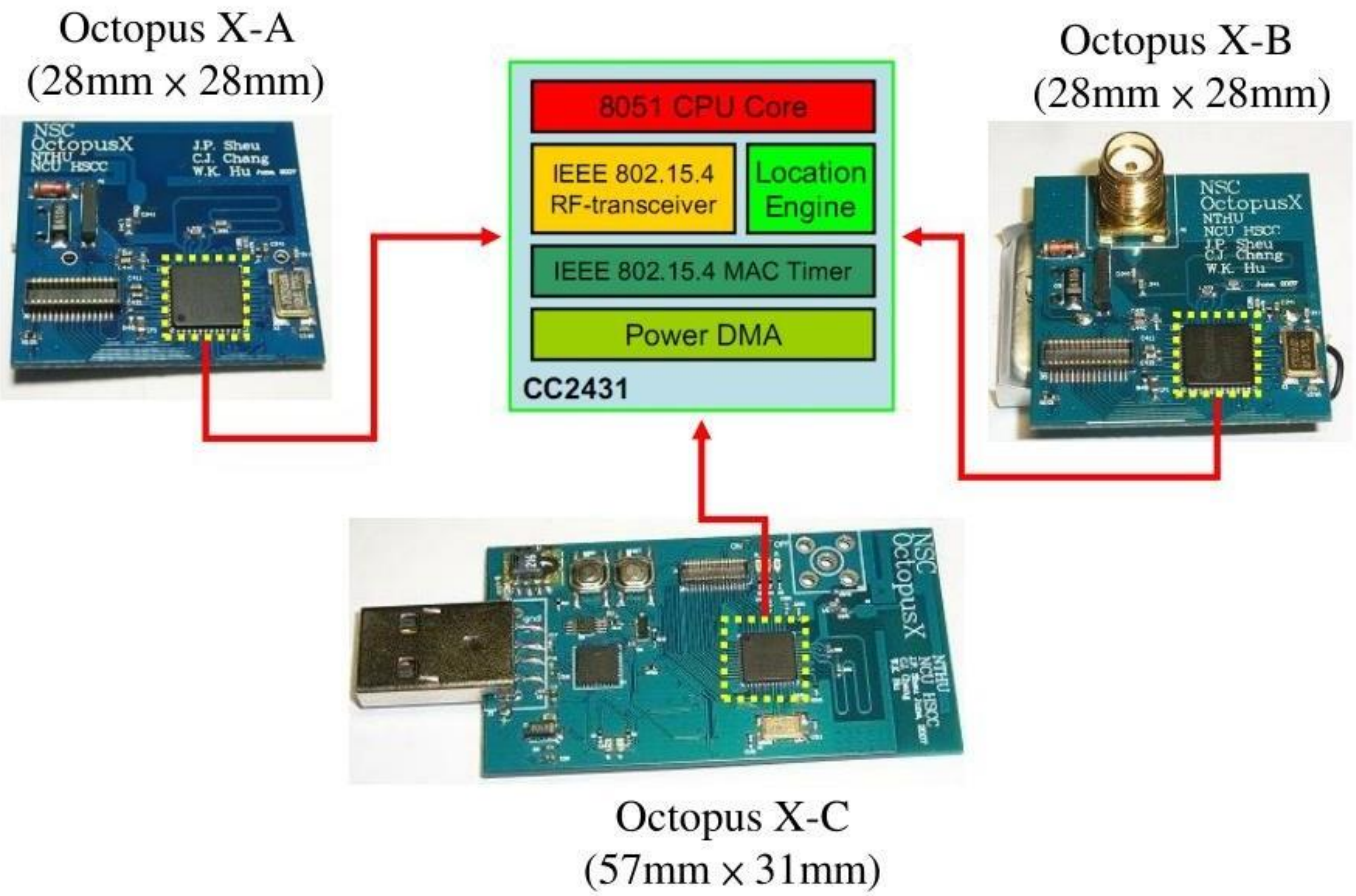
USB dongle



Temperature sensor

Three axis accelerometer

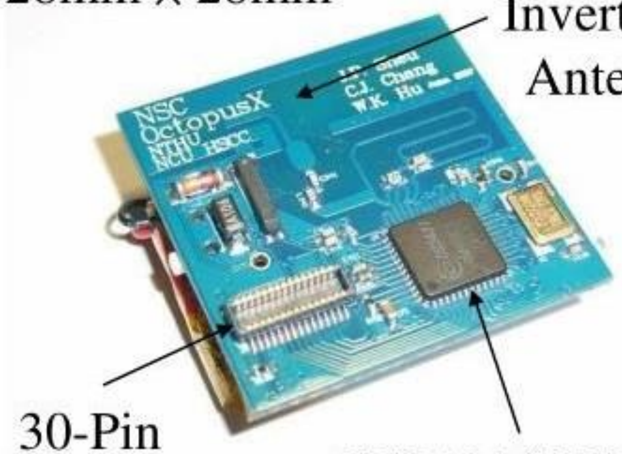
Introduction of Octopus X Hardware Platform



Features of Octopus X-A

Size: 28mm × 28mm

Inverted-F Antenna



30-Pin expansion connector

CC2431(MCU+RF)

Height: 7mm

Polymer battery

- MCU (CC2431)
- Inverted-F antenna
- RF transmission range ≒ 100m
- External crystal (32MHz+32.768KHz)
- 30-Pin expansion connector
- Polymer batter (3.7V 300mAh)

Features of Octopus X-B

Size: 28mm × 28mm

SMA Type Antenna

30-Pin expansion connector

CC2431(MCU+RF)

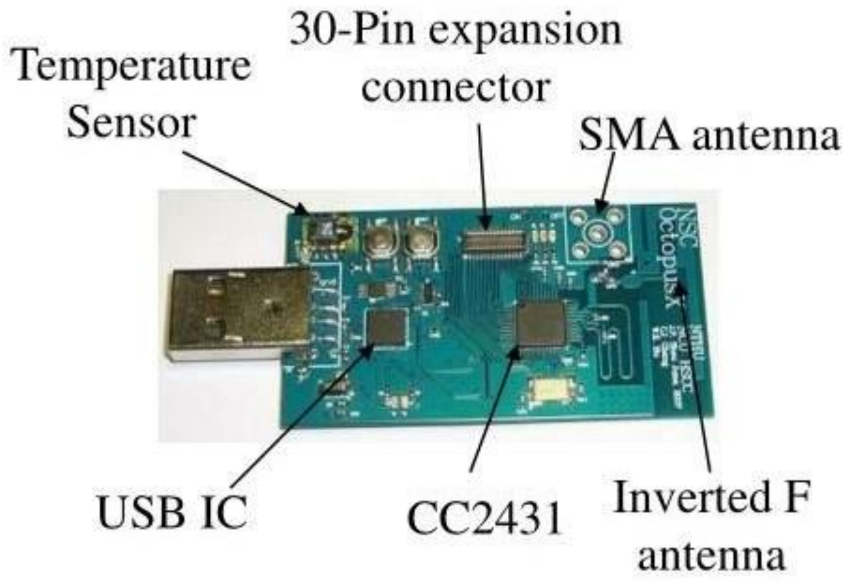
Height: 7mm

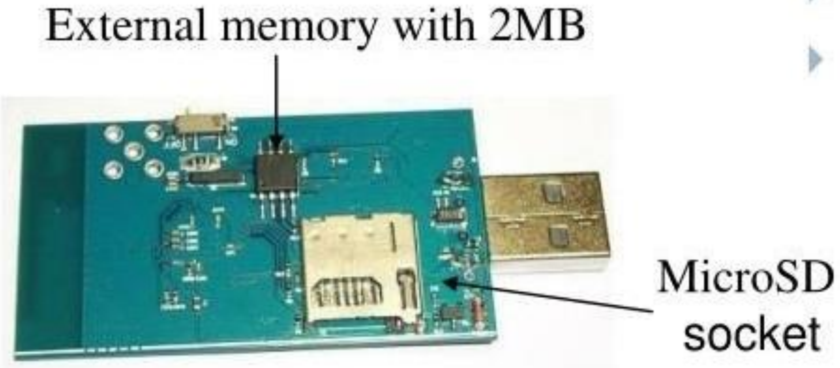
Polymer battery

- MCU (CC2431)
- SMA type antenna
- RF transmission range ≒ 150m
- External crystal (32MHz+32.768KHz)
- 30-Pin expansion connector
- Polymer batter (3.7V 300mAh)

Features of Octopus X-C

Size: 57mm × 31mm

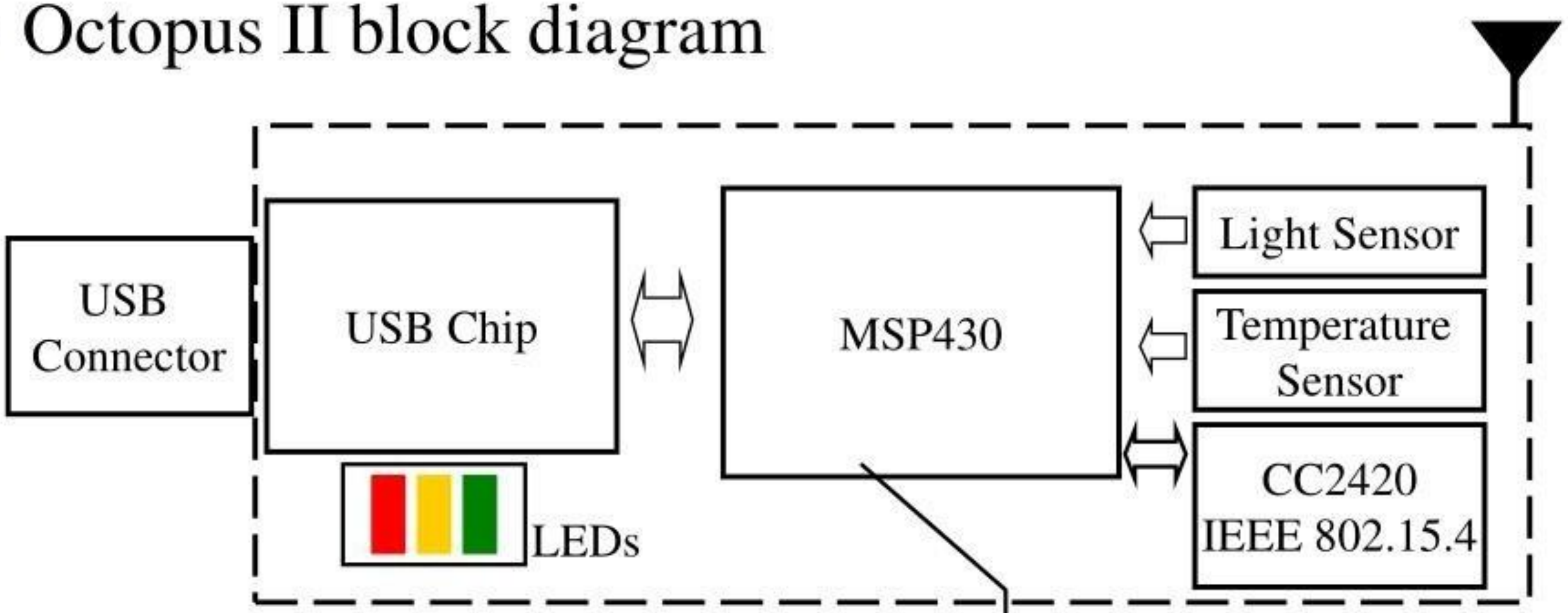




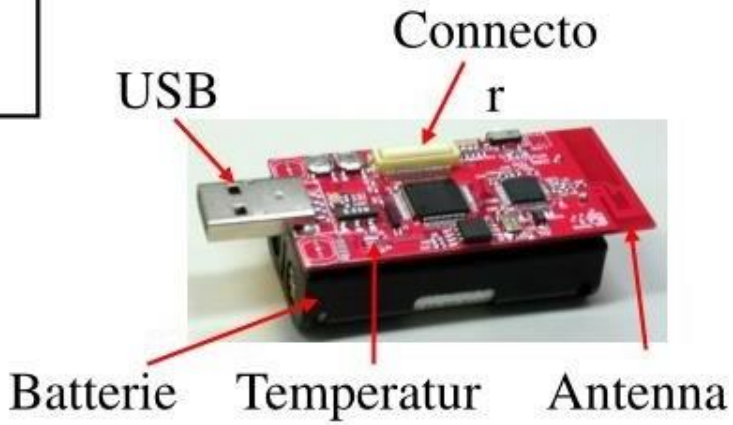
- MCU (CC2431)
- SMA type and Inverted-F antenna
- Humidity & Temperature sensor
  - Humidity 0~100%RH (0.03%RH)
  - Temperature -40°C~120°C (0.01°C)
- External flash memory (2MB)
- MicroSD socket (up to 8GB)
- USB Interface
  - Programming
  - Debugging
  - Data collection

Introduction of Octopus II Hardware Platform

- Octopus II block diagram



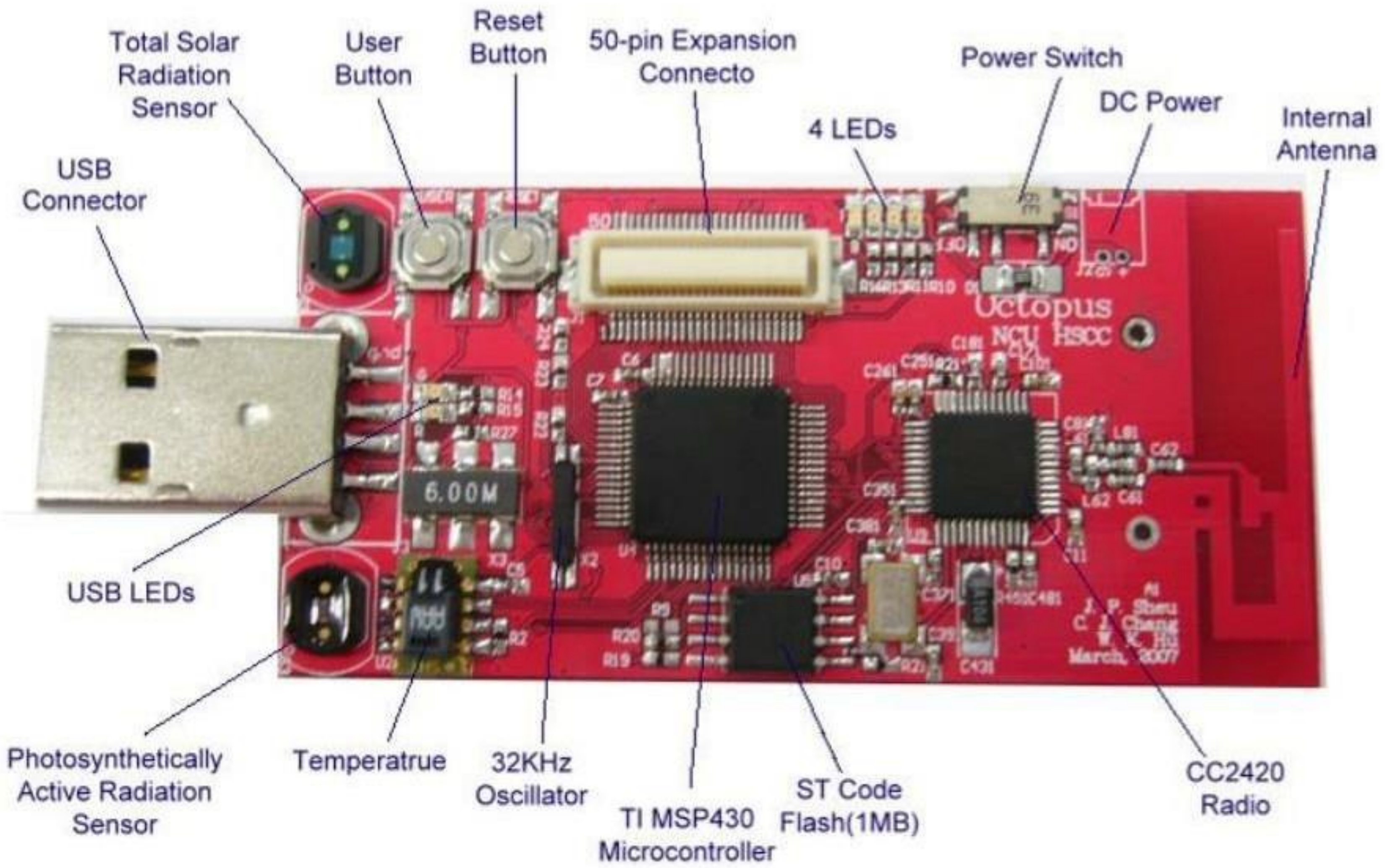
16-bit MSP430 microcontroller core 8MHz
48 KB in-system programmable flash
10 KB RAM
ADC 12-Bit 8 Channels



Features of Octopus II-A

- Front view of Octopus II-A

Size: 65mm × 31mm



Energy Consumption of Sensor Node

The Main Consumers of Energy

- Microcontroller
- Radio front ends
  - RF transceiver IC
  - RF antenna
- Degree of Memory
  - RAM
  - EEPROM
  - Flash memory
- Depending on the type of sensors
  - Temperature sensor
  - Humidity sensor
- Other components
  - LED
  - External Crystal
  - USB IC

Energy Consumption of Sensor Node

- A "back of the envelope" estimation for energy consumption
  - It means "energy consumption" is easily to estimate
- Number of instructions
  - Energy per instruction: 1 nJ [4]
  - Small battery ("smart dust"): 1 J = 1 Ws
  - Corresponds: 10^9 instructions!
- Lifetime
  - Require a single day operational lifetime
    = 24hr × 60mins × 60secs = 86400 secs
  - 1 Ws / 86400s ≒ **11.5 μW** as max. sustained power consumption!
- Not feasible!
  - Most of the time a wireless sensor node has nothing to do
  - Hence, it is best to turn it off

Multiple Power Consumption Modes

- Way out: Do not run sensor node at full operation all the time
  - If nothing to do, switch to ***power safe mode***
  - Question: When to throttle down? How to wake up again?
- Typical modes
  - Microcontroller
    - Active, Idle, Sleep
  - Radio mode
    - Turn on/off transmitter/receiver
- Multiple modes possible, "deeper" sleep modes
  - Strongly depends on hardware
  - Ex: TI MSP 430
    - Four different sleep modes
  - Atmel ATMega
    - Six different modes

Some Energy Consumption Figures

- Microcontroller power consumption
  - TI MSP 430 (@ 1 MHz, 3V) [6]
    - Fully operation : 1.2 mW
    - Deepest sleep mode : 0.3 μW
      - ✓ Only wake up by external interrupts (not even timer is running any more)
  - Atmel ATMega128L [7]
    - Operational mode:
      - ✓ Active : 15 mW
      - ✓ Idle : 6 mW
    - Sleep mode : 75 μW

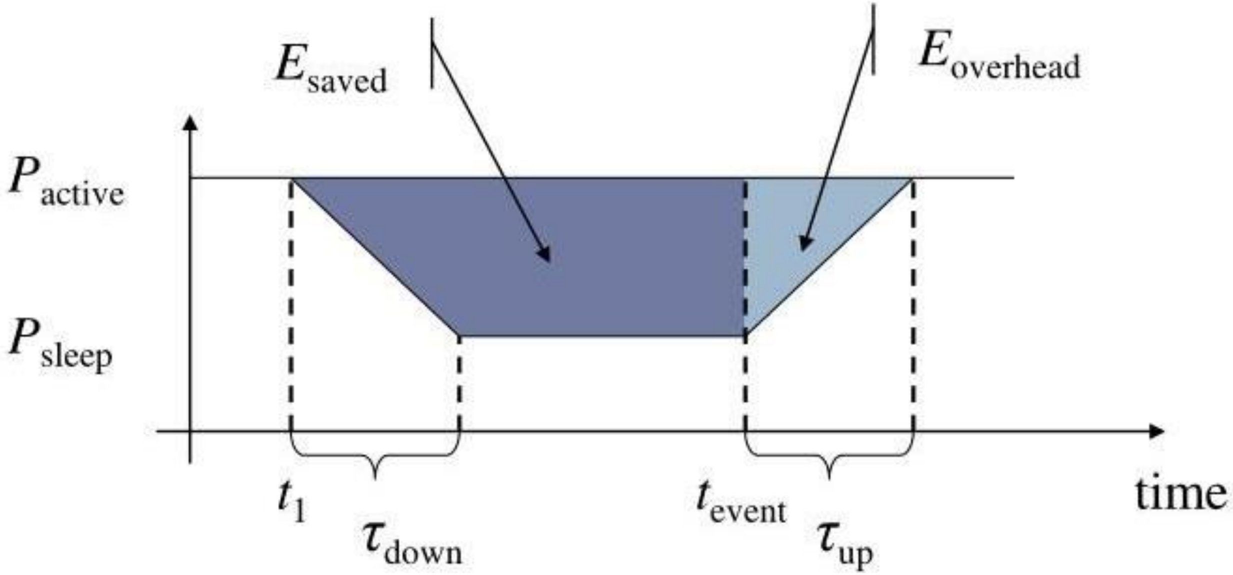
Some Energy Consumption Figures

- Memory power consumption
  - Power for RAM almost negligible
  - FLASH memory is crucial part

  - FLASH writing/erasing is expensive
    - Example: FLASH on Mica motes
    - Reading: $\risingdotseq 1.1\ nAh$ per byte
    - Writing: $\risingdotseq 83.3\ nAh$ per byte

Switching Between Modes

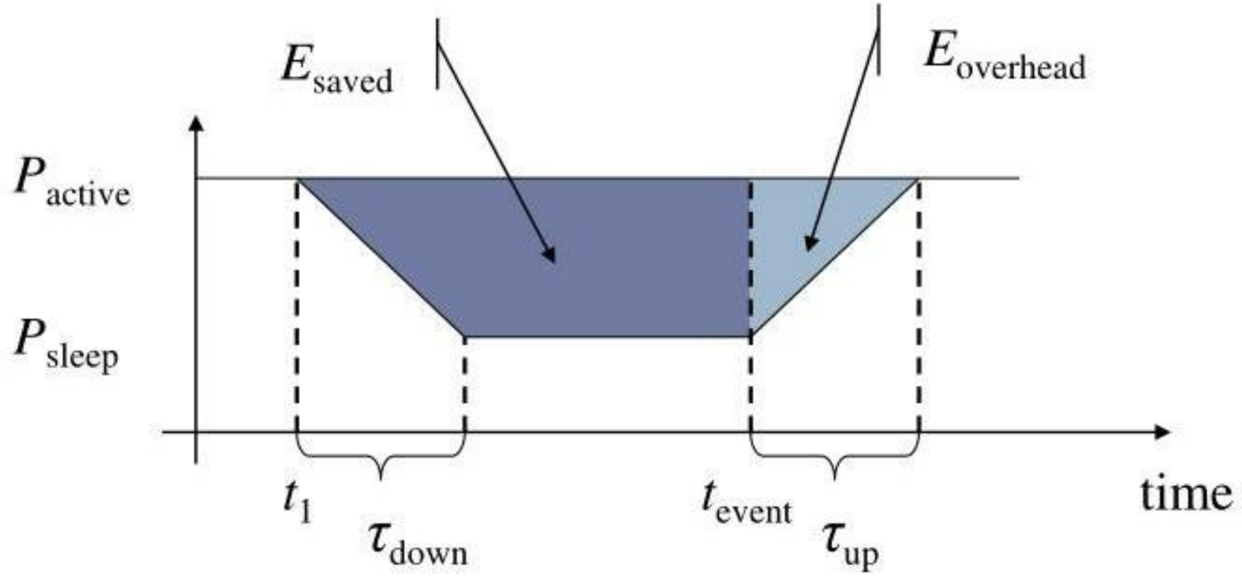
- Simplest idea: Greedily switch to lower mode whenever possible
- Problem: Time and power consumption required to reach higher modes not negligible
  - Introduces overhead
  - Switching only pays off if $E_{saved} > E_{overhead}$
- Example: Event-triggered wake up from sleep mode
- Scheduling problem with uncertainty



Switching Between Modes

- $E_{\text{saved}} = (t_{\text{event}} - t_1) \times P_{\text{active}} - (\tau_{\text{down}} \times (P_{\text{active}} + P_{\text{sleep}}) / 2 + (t_{\text{event}} - t_1 - \tau_{\text{down}}) \times P_{\text{sleep}})$

- $E_{\text{overhead}} = \tau_{\text{up}} \times (P_{\text{active}} - P_{\text{sleep}}) / 2$



Power Consumption vs. Transmission Distance

- Free space loss: direct-path signal

$$P_r = P_t G_r G_t \frac{\lambda^2}{(4\pi)^2 (d)^2} = \frac{A_r A_t}{(\lambda d)^2}$$

  - d = distance between transmitter and receiver
  - P_t = transmitting power
  - P_r = receiving power
  - G_t = gain of transmitting antenna
  - G_r = gain of receiving antenna
  - A_t = effective area of transmitting antenna
  - A_r = effective area of receiving antenna

Power Consumption vs. Transmission Distance

- Two-path model

$$P_r = P_t G_r G_t \left(\frac{h_t h_r}{d^2}\right)^2$$

- h_t and h_r are the height of the transmitter and receiver

- The general form

$$P_r = P_t G_r G_t \left(\frac{\lambda}{4\pi}\right)^2 \frac{1}{d^\gamma}$$

  - γ is the propagation coefficient that varies 2 ~ 5

Computation vs. Communication Energy Cost

- Tradeoff ?
  - It's not possible to directly compare computation/communication energy cost
  - Energy ratio of "sending one bit" vs. "computing one instruction"
  - Communication (send & receive) 1 KB ≒ Computing 3,000,000 (3 million) instructions [10]
- Hence
  - Try to compute instead of communication whenever possible
- Key technique in WSN
  - In-network processing
  - Exploit data centric/aggregation, data compression, intelligent coding, signal processing …

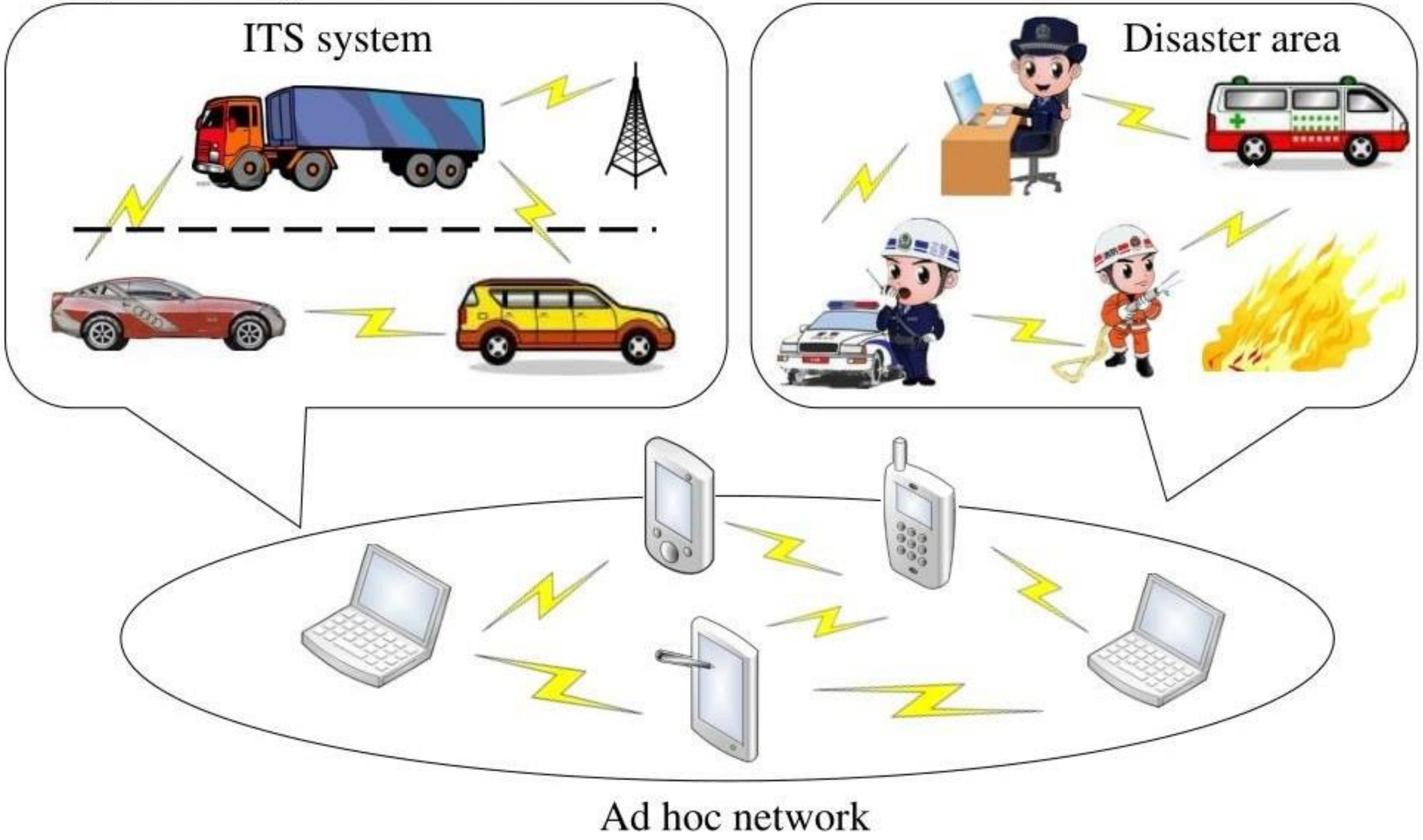
2.4. Network Architecture

Difference between Ad hoc and Sensor Networks

- (Mobile) Ad hoc Scenarios
  - Nodes communicate with each other
    - That means each node can be a source node or destination node
  - Nodes can communicate "some" node in another network
    - Ex: Access to Web/Mail/DNS server on the Internet
    - Typically requires some connection to the fixed network
- Applications of Ad hoc network
  - Traditional data (http, ftp, collaborative apps, …)
  - Multimedia (voice, video)

Difference between Ad hoc and Sensor Networks

- (Mobile) Ad hoc Scenarios

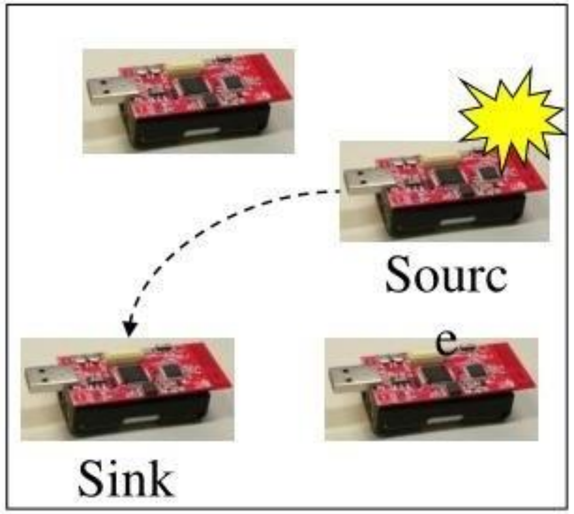


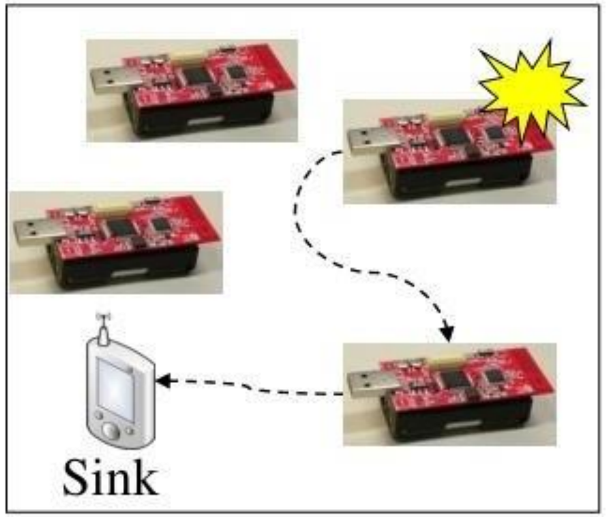
Difference between Ad hoc and Sensor Networks

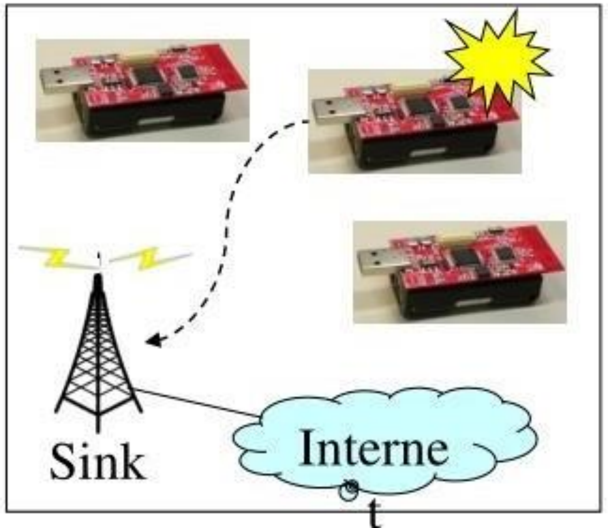
- Sensor Network Scenarios
  - **Sources**: Any sensor node that provides sensing data/measurements
  - **Sinks**: Sensor nodes where information is required
    - Belongs to the sensor network
    - Could be the same sensor node or an external entity such mobile phone/NB/Table PC
    - Is part of an external network (e.g., internet), somehow connected to the WSN
- Applications of Sensor Network
  - Usually, machine to machine
  - Often limited amounts of data
  - Many different kinds of applications

Difference between Ad hoc and Sensor Networks

- Sensor Network Scenarios



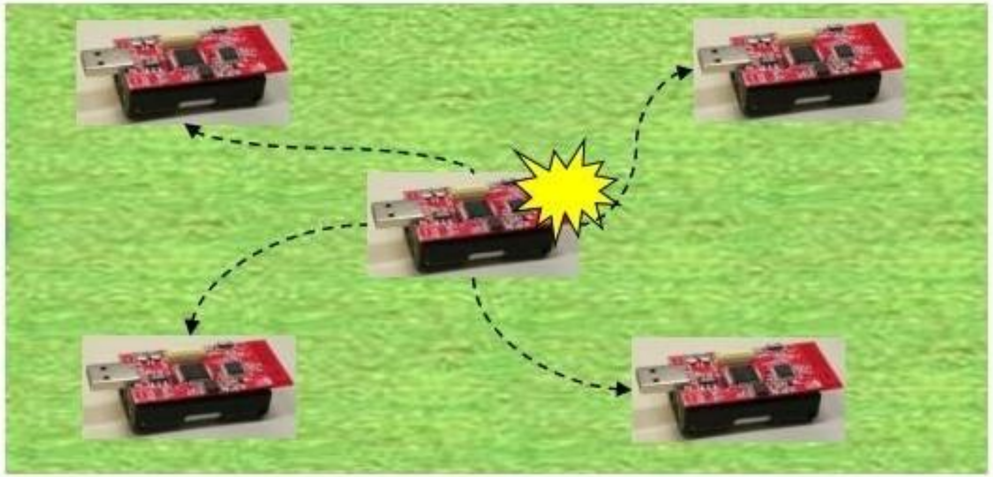




Single-hop vs. Multi-hop Networks

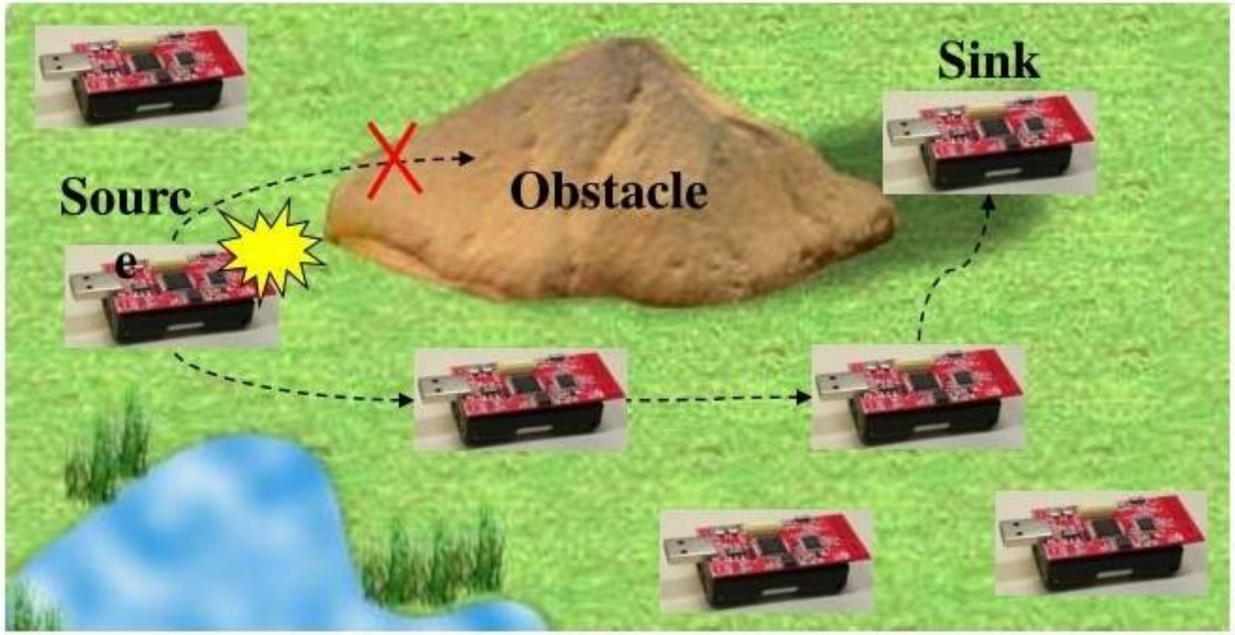
- One common problem: limited range of wireless communication
  - Limited transmission power
  - Path loss
  - Obstacles
- Solution: multi-hop networks
  - Send packets to an intermediate node
  - Intermediate node forwards packet to its destination
  - **Store-and-forward** multi-hop network
- Basic technique applies to both WSN and MANET
- Note:
  - Store-and-forward multi-hopping NOT the only possible solution
    - Ex: Collaborative networking, Network coding [11] [12]….

Single-hop vs. Multi-hop Networks

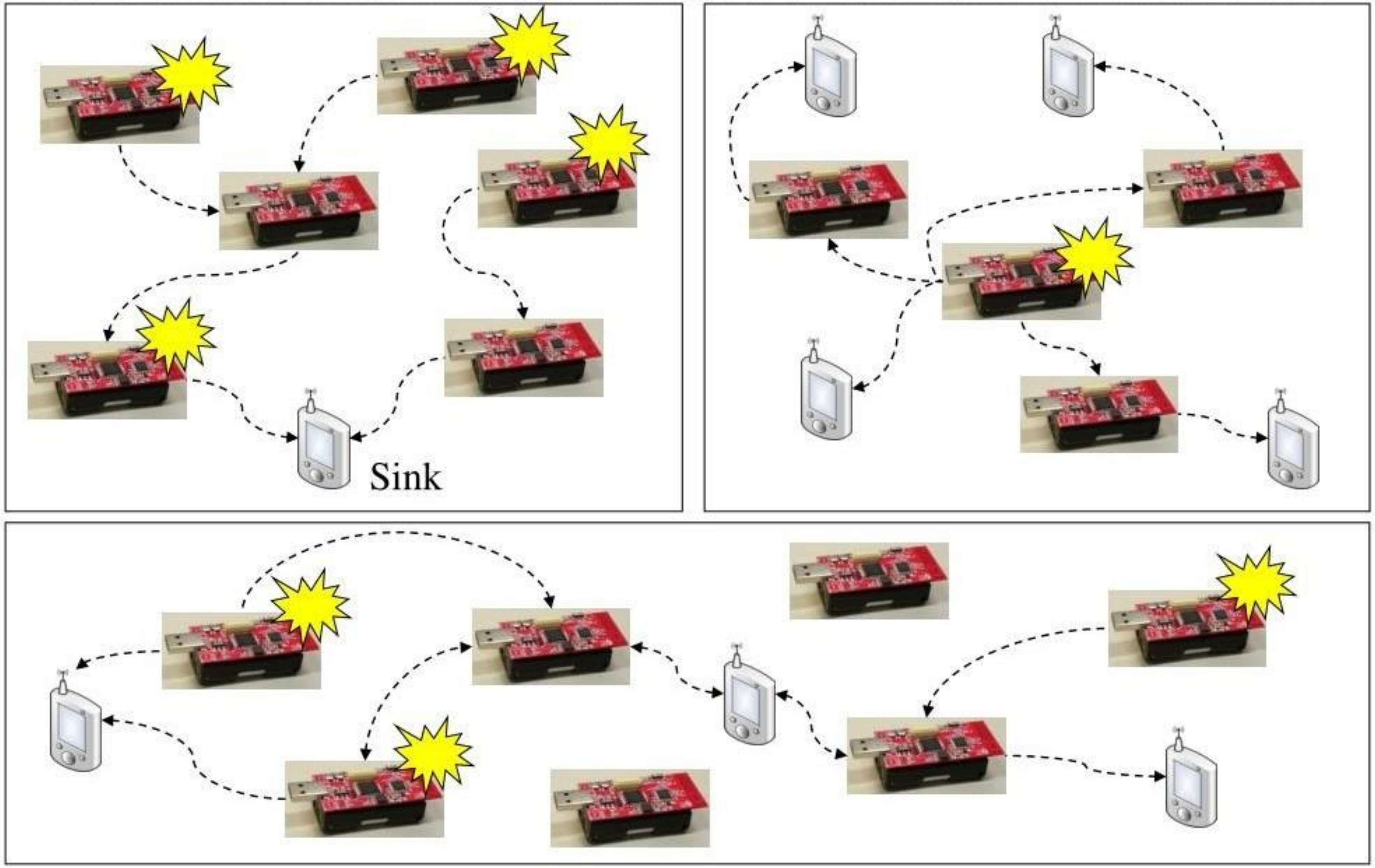


Single-hop networks

Multi-hop networks



Multiple Sinks, Multiple Sources WSN



In-network Processing

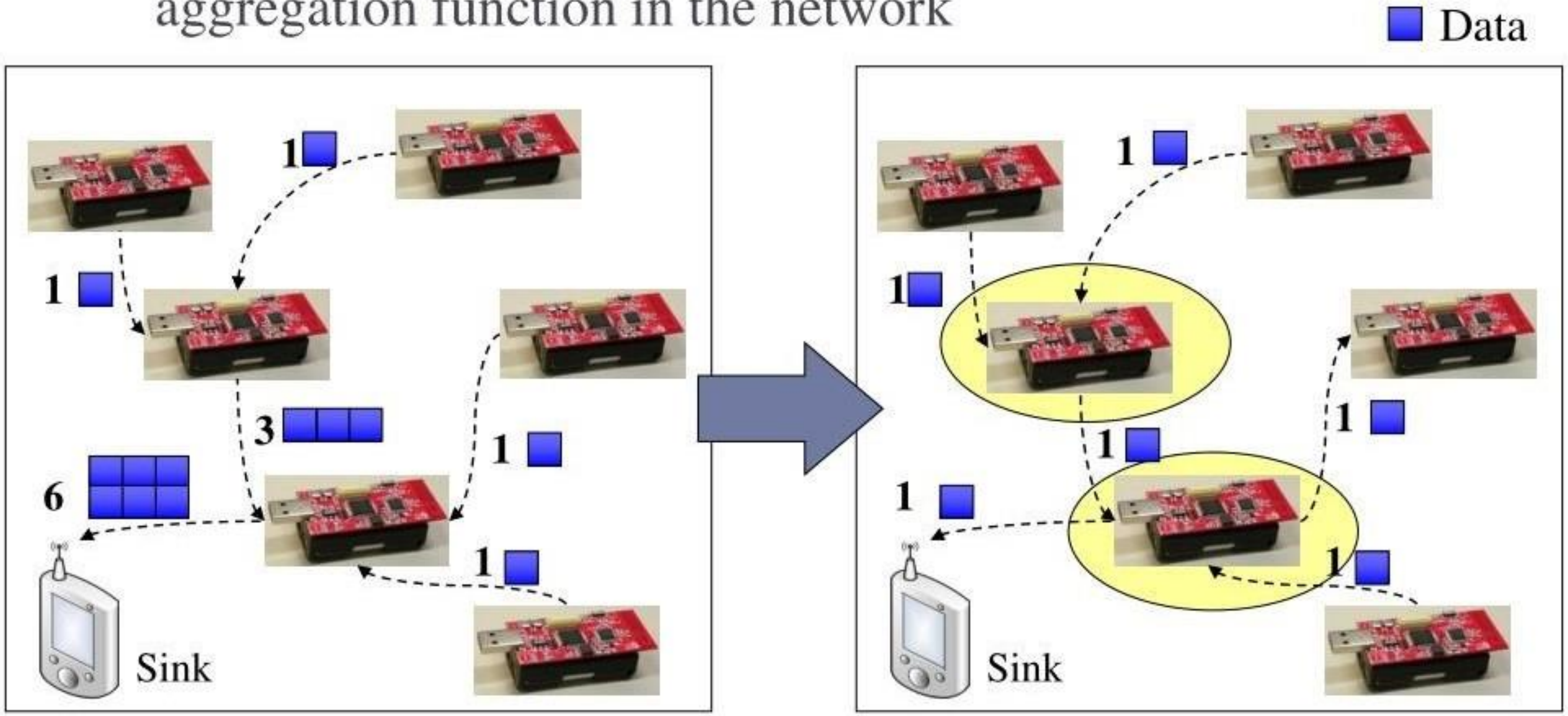
- MANETs are supposed to deliver bits from one end to the other

- WSNs, on the other end, are expected to provide information, not necessarily original bits
  - Ex: ***manipulate*** or ***process*** the data in the network

- Main example: aggregation
  - Apply composable **[13]** aggregation functions to a convergecast tree in a network
  - Typical functions: minimum, maximum, average, sum, …

In-network Processing

- Processing Aggregation example
  - The simplest in-network processing technique
  - Reduce number of transmitted bits/packets by applying an aggregation function in the network

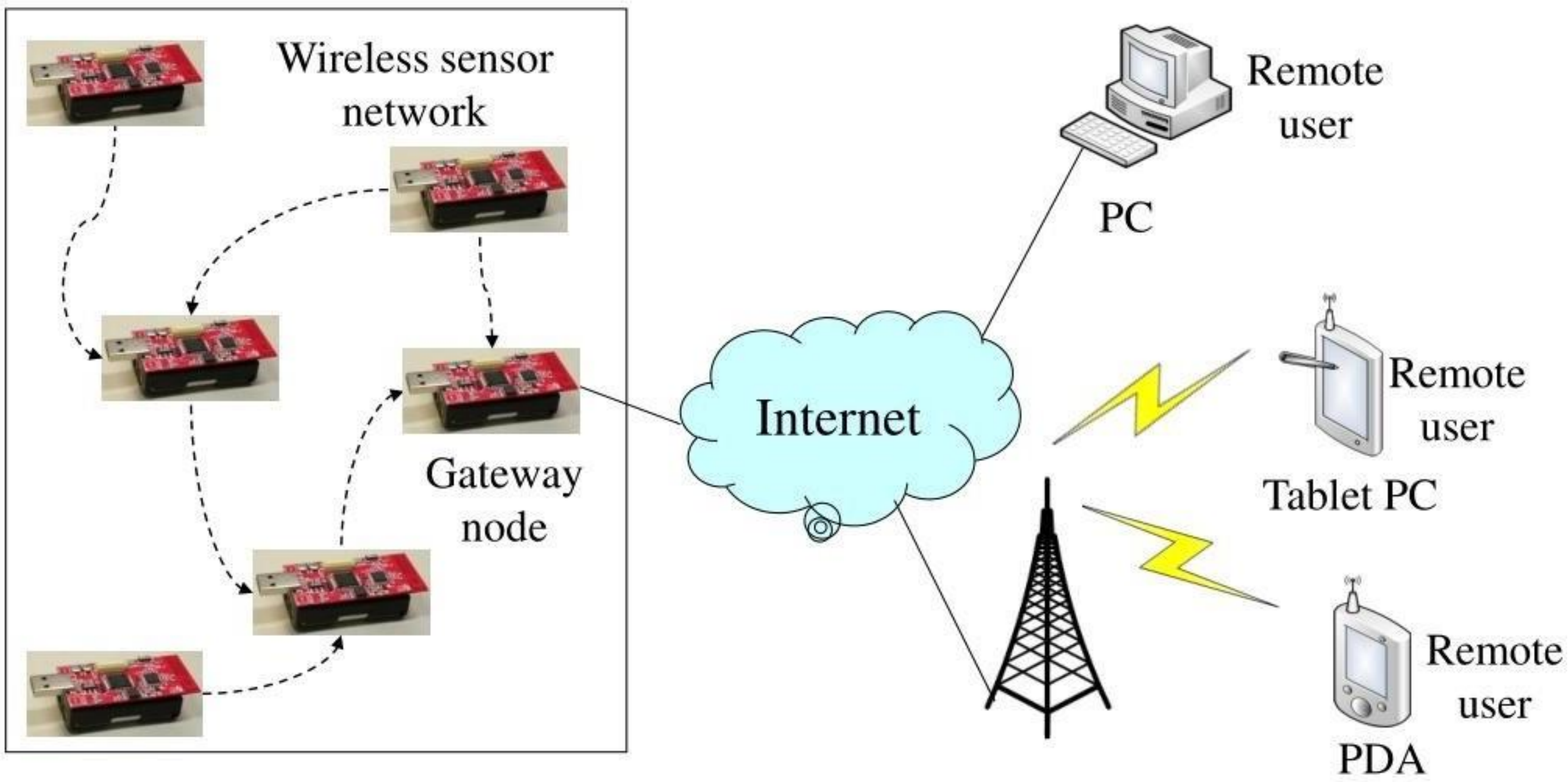


Gateway Concepts for WSN/MANET

- Gateways are necessary to the Internet for remote access to/from the WSN
  - For ad hoc networks
    - Additional complications due to mobility
      - ✓ Ex: Change route to the gateway, use different gateways
  - For WSN
    - Additionally bridge the gap between different interaction semantics in the gateway

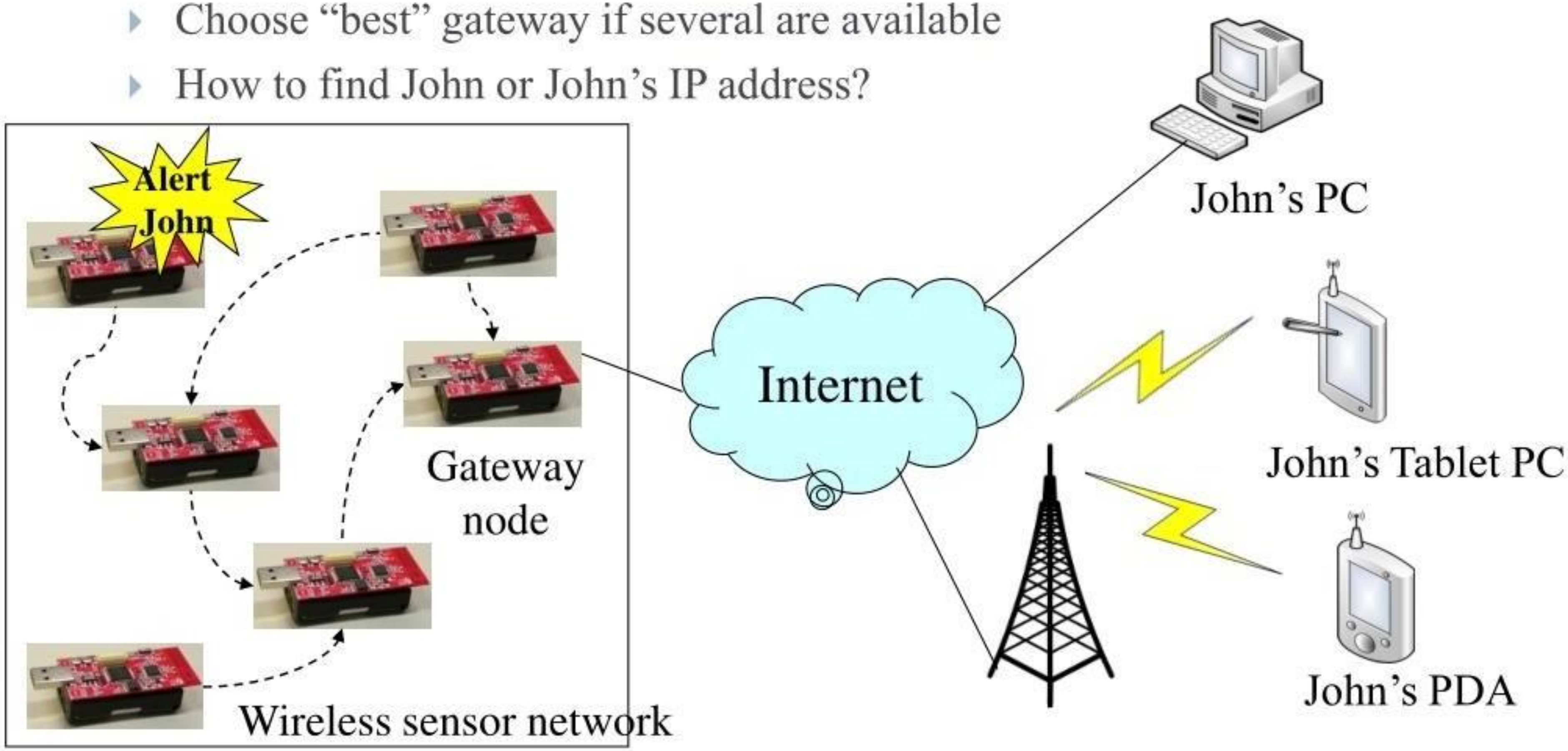
Gateway Concepts for WSN/MANET

- Gateway support for different radios/protocols, …



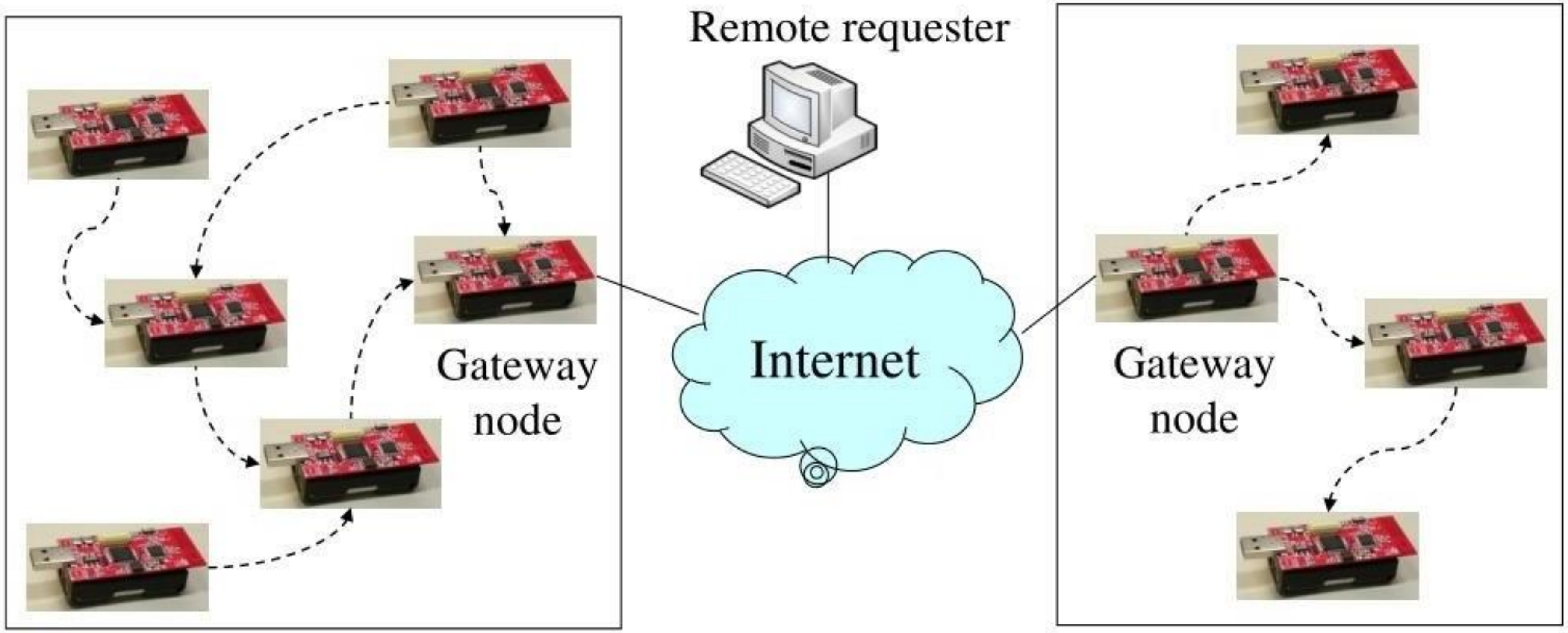
WSN to Internet Communications

- Scenario: Deliver an alarm message to an Internet host
- Problems
  - Need to find a gateway (integrates routing & service discovery)
  - Choose “best” gateway if several are available
  - How to find John or John’s IP address?



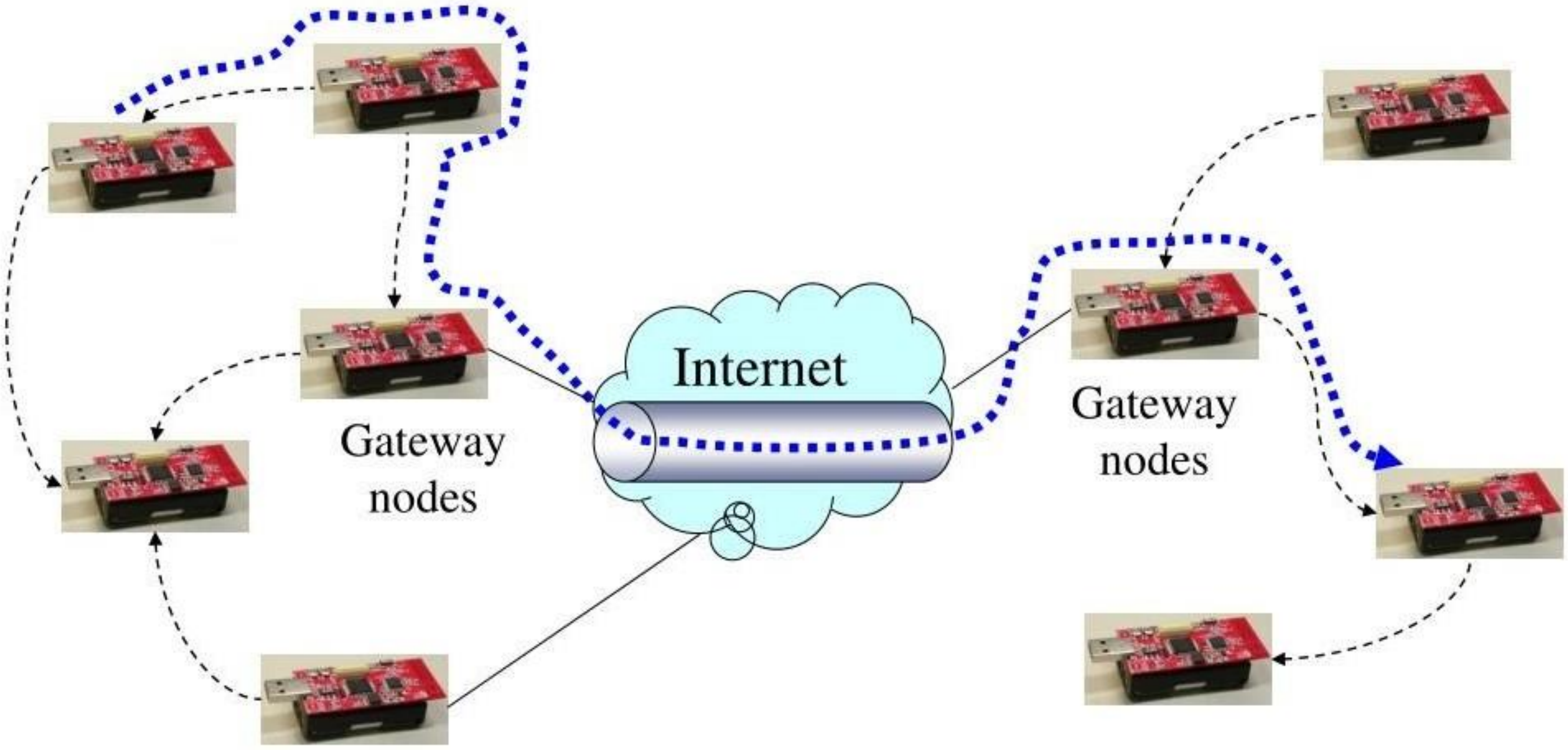
Internet to WSN communications

- How to find the right WSN to answer a need?
- How to translate from IP protocols to WSN protocols, semantics?



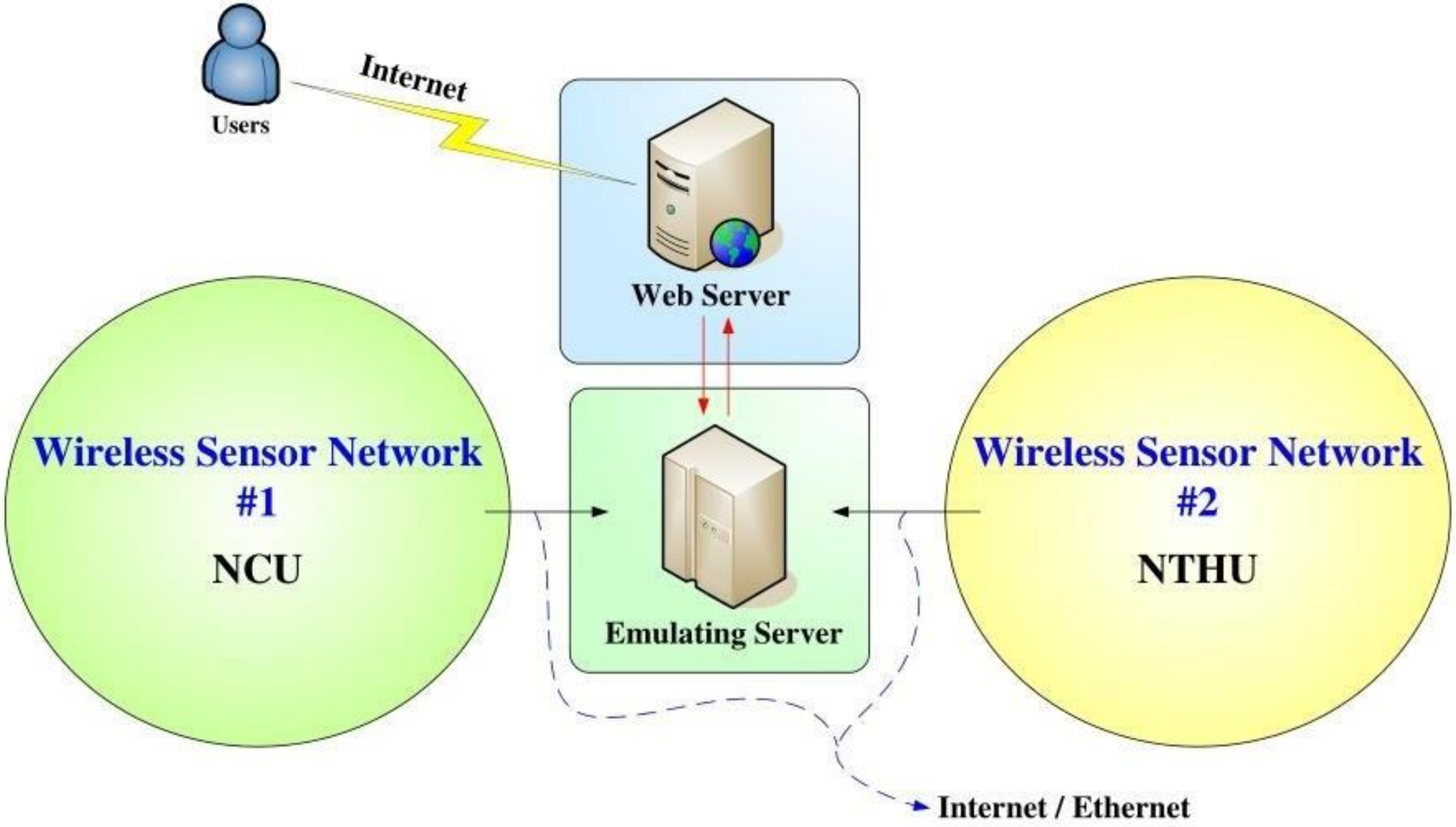
WSN Tunneling

- The idea is to build a larger, "Virtual" WSN
- Use the Internet to "tunnel" WSN packets between two remote WSNs



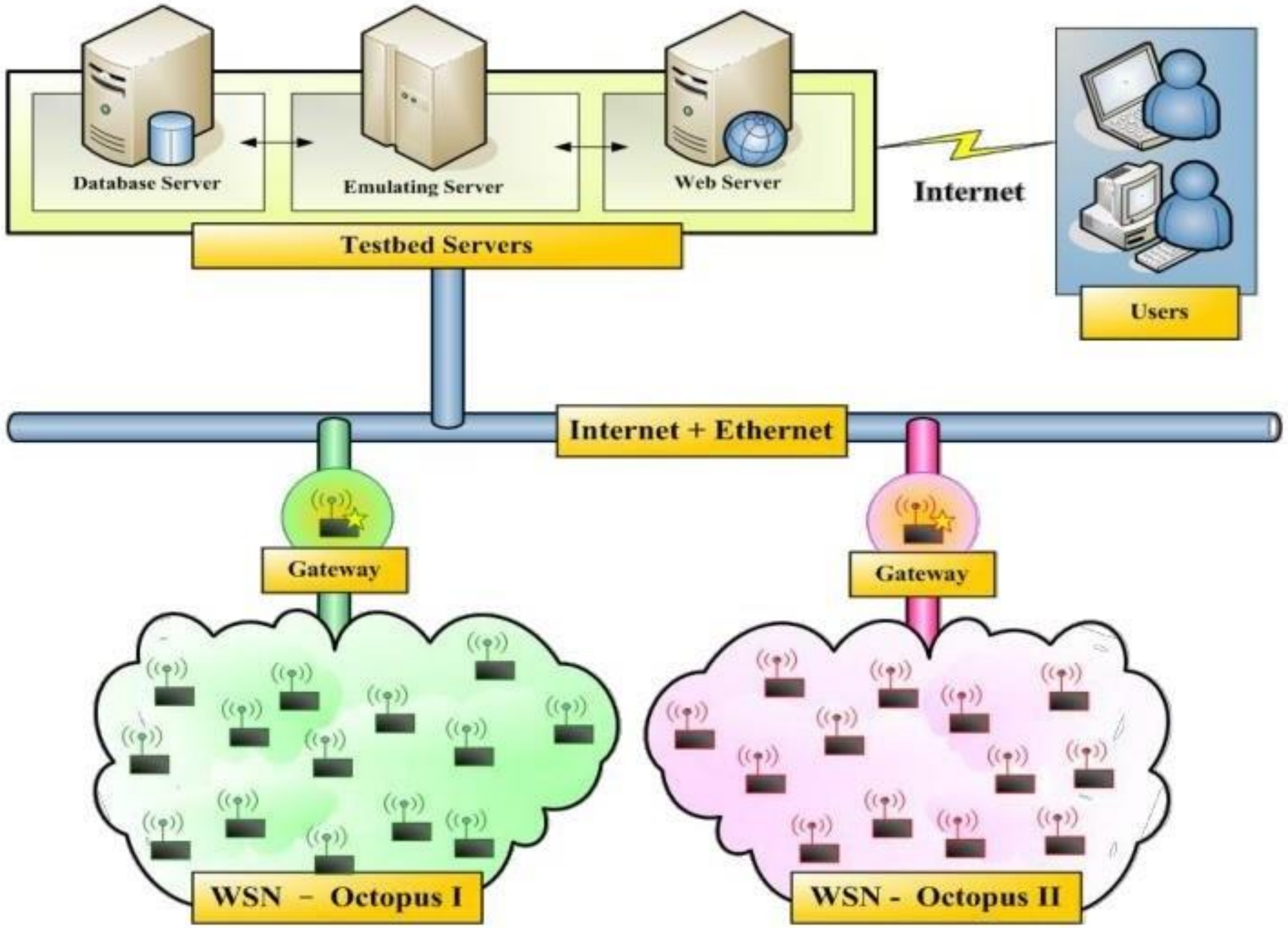
WSN Tunneling

- Example of WSN tunneling
  - WSNs Testbed



WSN Tunneling

- Example of WSN tunneling
  - Testbed scenario



2.5. Challenges of Sensor Nodes

Challenges of Wireless Sensor Node

- More energy-efficient
  - Self-sufficiency in power supply such as the installation of solar collector panels
  - Design more energy-efficient of the circuit, or to adopt more energy-efficient electronic components
- Integrating more sensors
  - For multiple purposes such as detecting human's motion, temperature, blood pressure and heartbeat at the same time
- Higher processing performance
  - In future, more complex application need more powerful computation

Challenges of Wireless Sensor Node

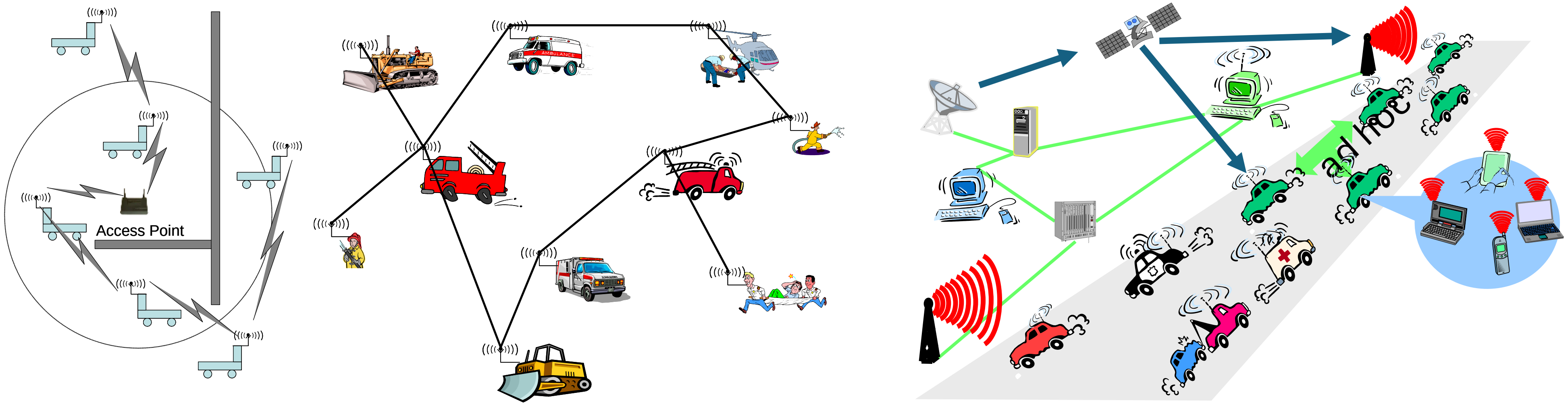
- More Robust and Secure
  - Not easy damaged or be destroyed
  - Secure transmission of sensing data and not easy being tapped
- Easy to buy and deployment
  - Low price and easy to use

Outline

- ***Network scenarios***
- Optimization goals
- Gateway concepts

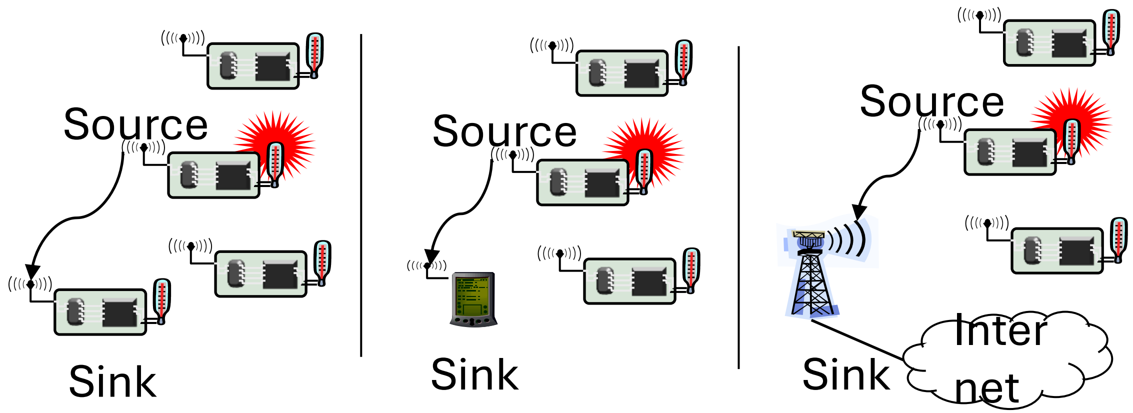
Basic scenarios: Ad hoc networks

- (Mobile) ad hoc scenarios
  - Nodes talking to each other
  - Nodes talking to “some” node in another network (Web server on the Internet, e.g.)
    - Typically requires some connection to the fixed network
  - Applications: Traditional data (http, ftp, collaborative apps, …) & multimedia (voice, video) ! humans in the loop



Basic scenarios: sensor networks

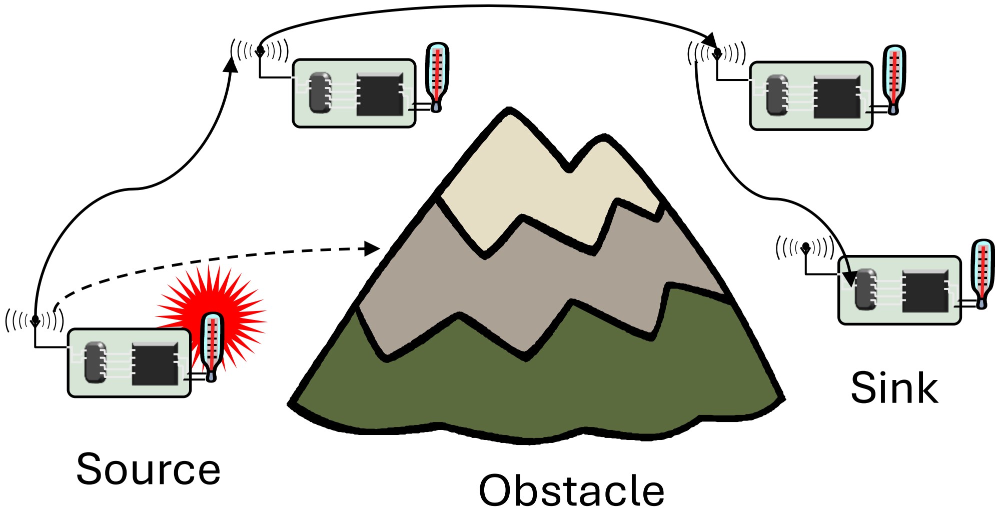
- Sensor network scenarios
  - ***Sources***: Any entity that provides data/measurements
  - ***Sinks***: Nodes where information is required
    - Belongs to the sensor network as such
    - Is an external entity, e.g., a PDA, but directly connected to the WSN
      - Main difference: comes and goes, often moves around, ...
    - Is part of an external network (e.g., internet), somehow connected to the WSN
  - Applications: Usually, machine to machine, often limited amounts of data, different notions of importance



Single-hop vs. multi-hop networks

- One common problem: limited range of wireless communication
  - Essentially due to limited transmission power, path loss, obstacles
- Option: multi-hop networks
  - Send packets to an intermediate node
  - Intermediate node forwards packet to its destination
  - ***Store-and-forward*** multi-hop network

- Basic technique applies to both WSN and MANET
- Note: Store&forward multi-hopping NOT the only possible solution
  - E.g., collaborative networking, network coding
  - Do not operate on a per-packet basis

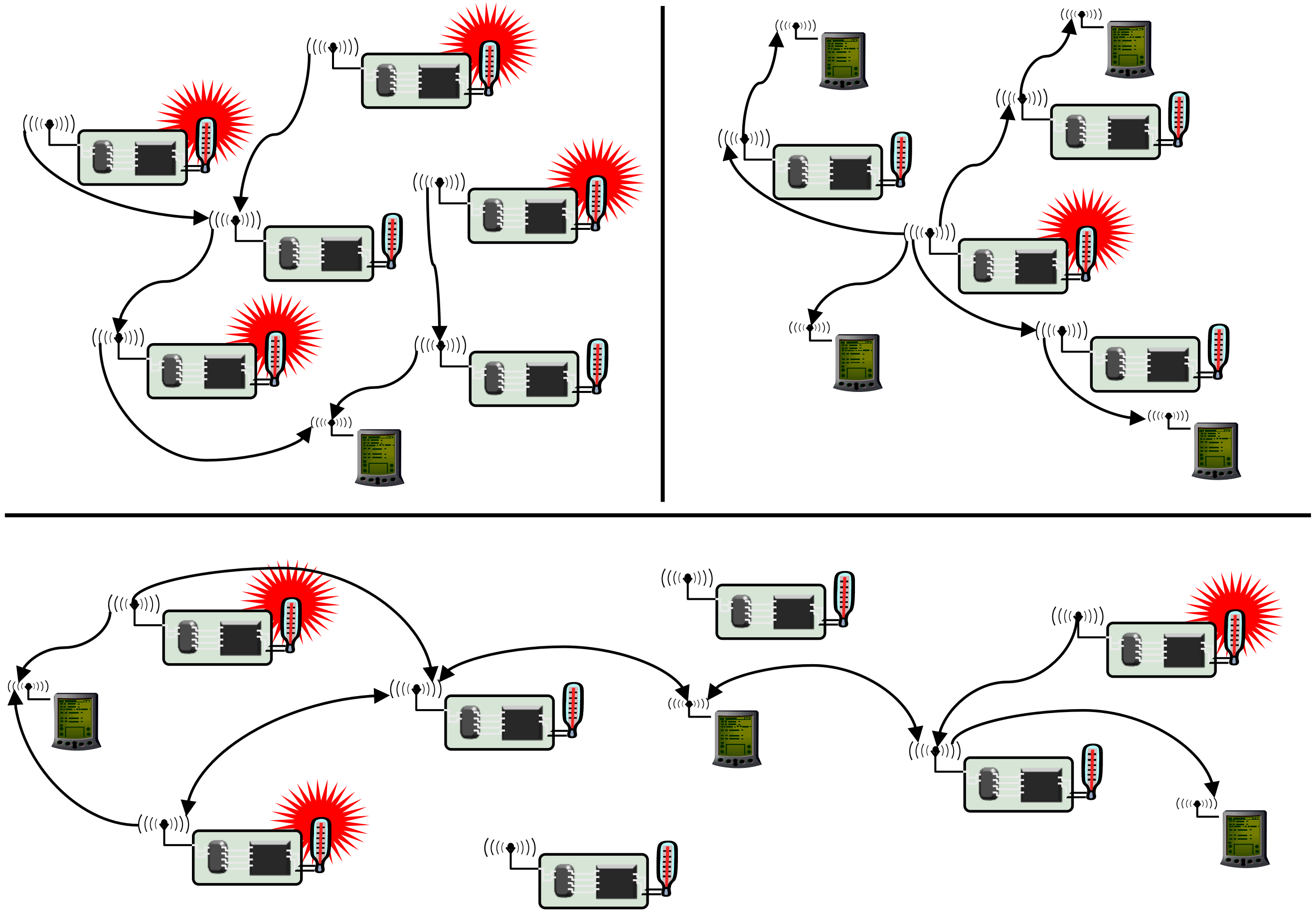


Energy efficiency of multi-hopping?

- Obvious idea: Multi-hopping is more energy-efficient than direct communication
  - Because of path loss $\alpha > 2$, energy for distance d is reduced from cd^α to $2c(d/2)^\alpha$
    - c some constant
- However: This is usually wrong, or at least very over-simplified
  - Need to take constant offsets for powering transmitter, receiver into account
  - Details see exercise, chapter 2

! Multi-hopping for energy savings needs careful choice

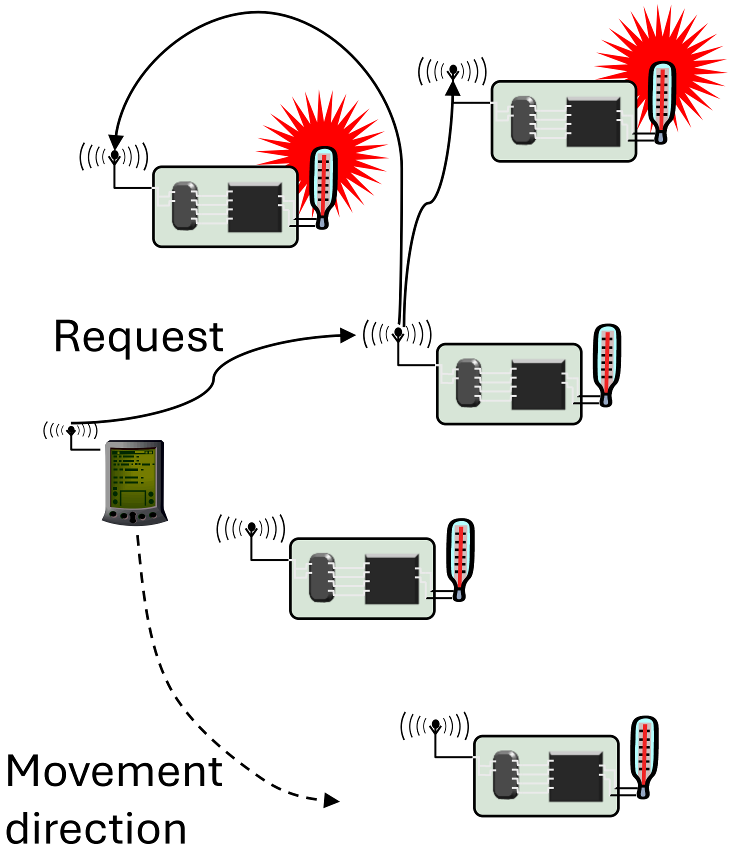
WSN: Multiple sinks, multiple sources

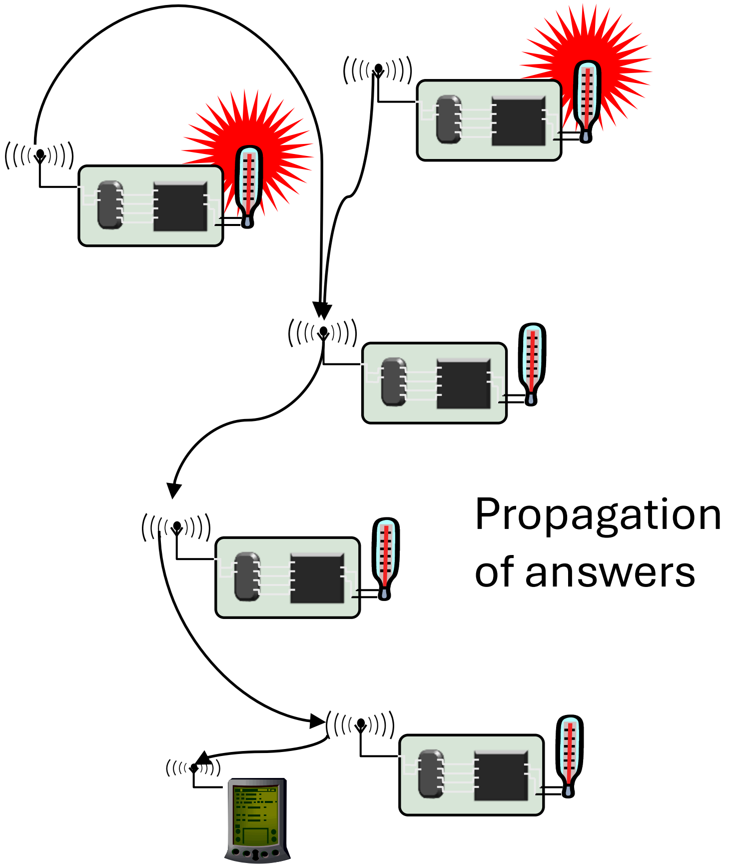


Different sources of mobility

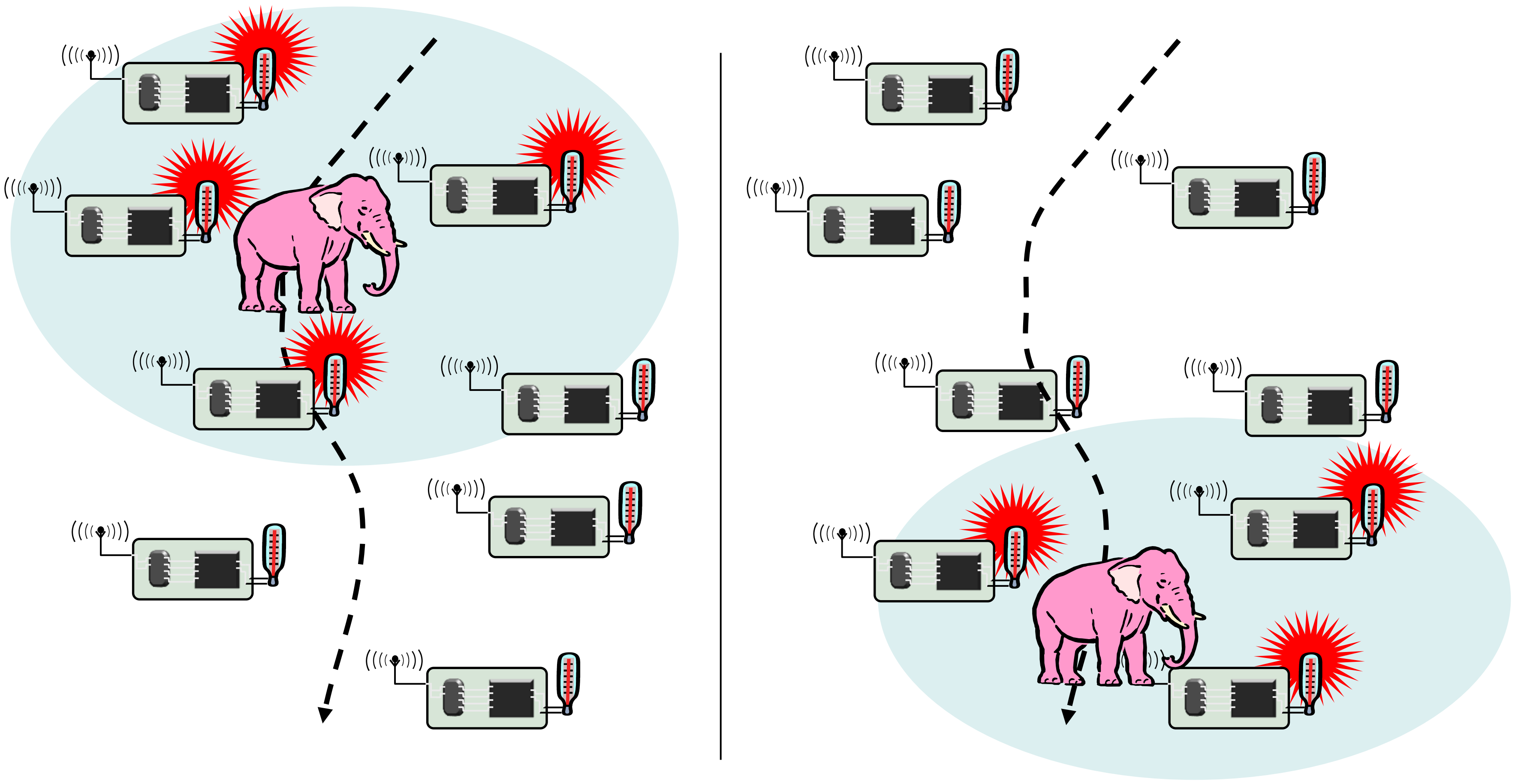
- Node mobility
  - A node participating as source/sink (or destination) or a relay node might move around
  - Deliberately, self-propelled or by external force; targeted or at random
  - Happens in both WSN and MANET
- Sink mobility
  - In WSN, a sink that is not part of the WSN might move
  - Mobile requester
- Event mobility
  - In WSN, event that is to be observed moves around (or extends, shrinks)
  - Different WSN nodes become “responsible” for surveillance of such an event

WSN sink mobility





WSN event mobility: Track the pink elephant



Here: Frisbee model as example

Optimization goal: Quality of Service

- In MANET: Usual QoS interpretation
  - Throughput/delay/jitter
  - High perceived QoS for multimedia applications
- In WSN, more complicated
  - Event detection/reporting probability
  - Event classification error, detection delay
  - Probability of missing a periodic report
  - Approximation accuracy (e.g, when WSN constructs a temperature map)
  - Tracking accuracy (e.g., difference between true and conjectured position of the pink elephant)
- Related goal: robustness
  - Network should withstand failure of some nodes

Optimization goal: Energy efficiency

- Umbrella term!
- Energy per correctly received bit
  - Counting all the overheads, in intermediate nodes, etc.
- Energy per reported (unique) event
  - After all, information is important, not payload bits!
  - Typical for WSN
- Delay/energy tradeoffs
- Network lifetime
  - Time to first node failure
  - Network half-life (how long until 50% of the nodes died?)
  - Time to partition
  - Time to loss of coverage
  - Time to failure of first event notification

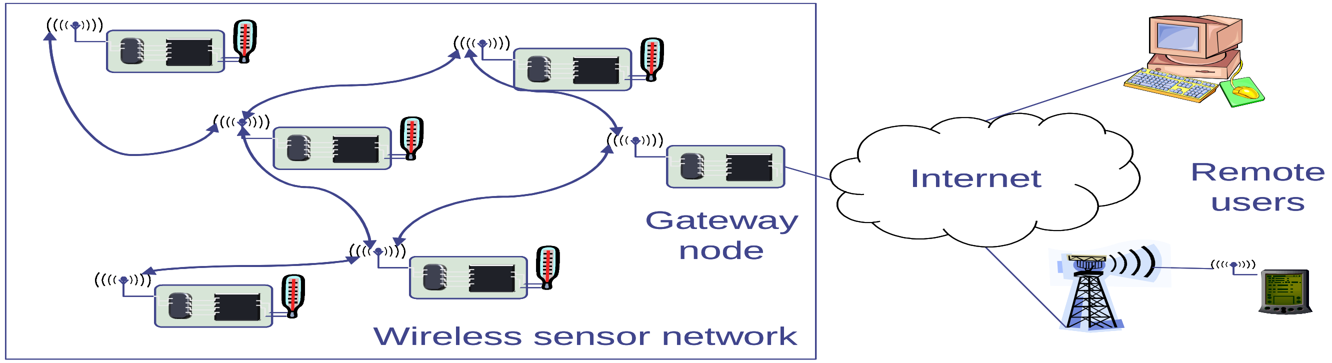
Optimization goal: Scalability

- Network should be operational regardless of number of nodes
  - At high efficiency
- Typical node numbers difficult to guess
  - MANETs: 10s to 100s
  - WSNs: 10s to 1000s, maybe more (although few people have seen such a network before…)

- Requiring to scale to large node numbers has ***serious*** consequences for network architecture
  - Might not result in the most efficient solutions for small networks!
  - Carefully consider actual application needs before looking for n ! 1 solutions!

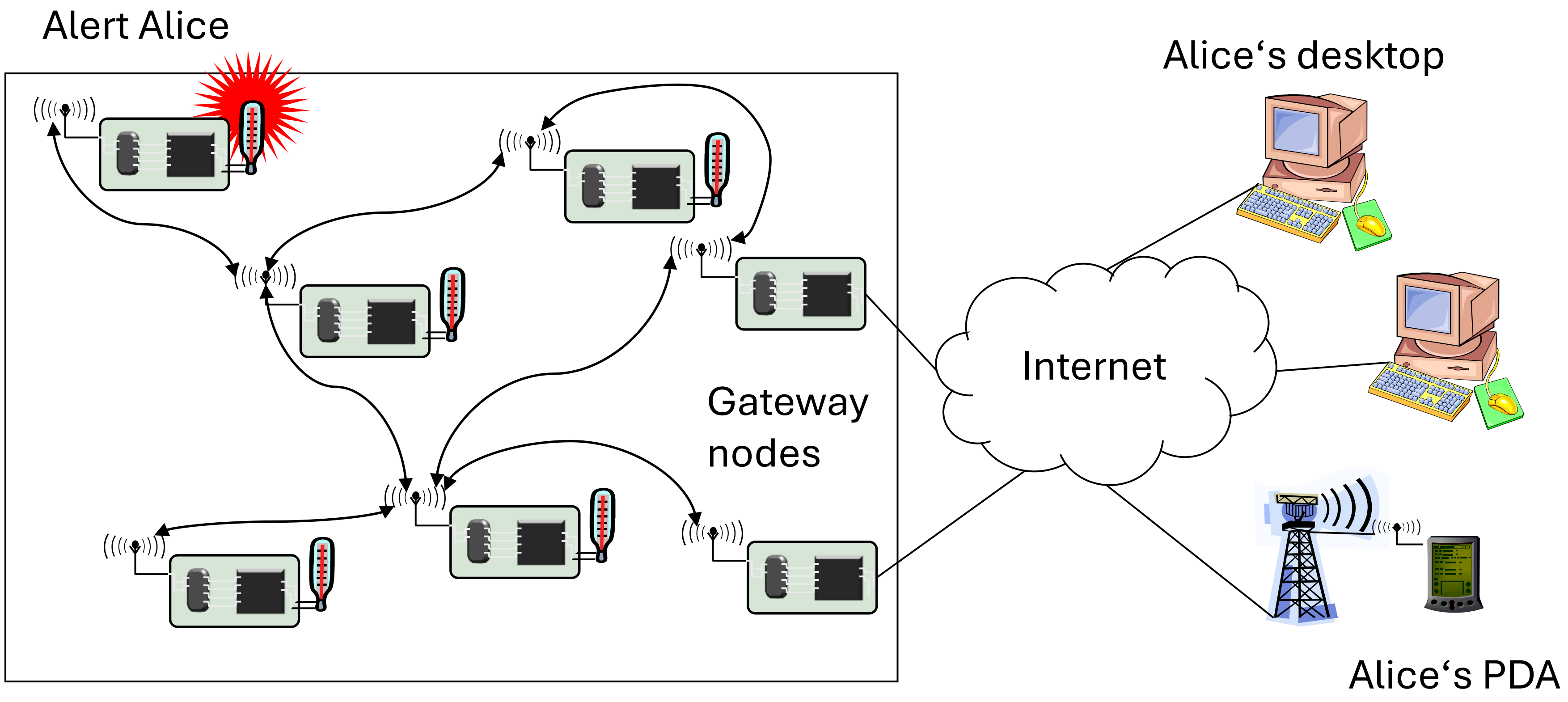
Gateway concepts for WSN/MANET

- Gateways are necessary to the Internet for remote access to/from the WSN
  - Same is true for ad hoc networks; additional complications due to mobility (change route to the gateway; use different gateways)
  - WSN: Additionally bridge the gap between different interaction semantics (data vs. address-centric networking) in the gateway
- Gateway needs support for different radios/protocols, …



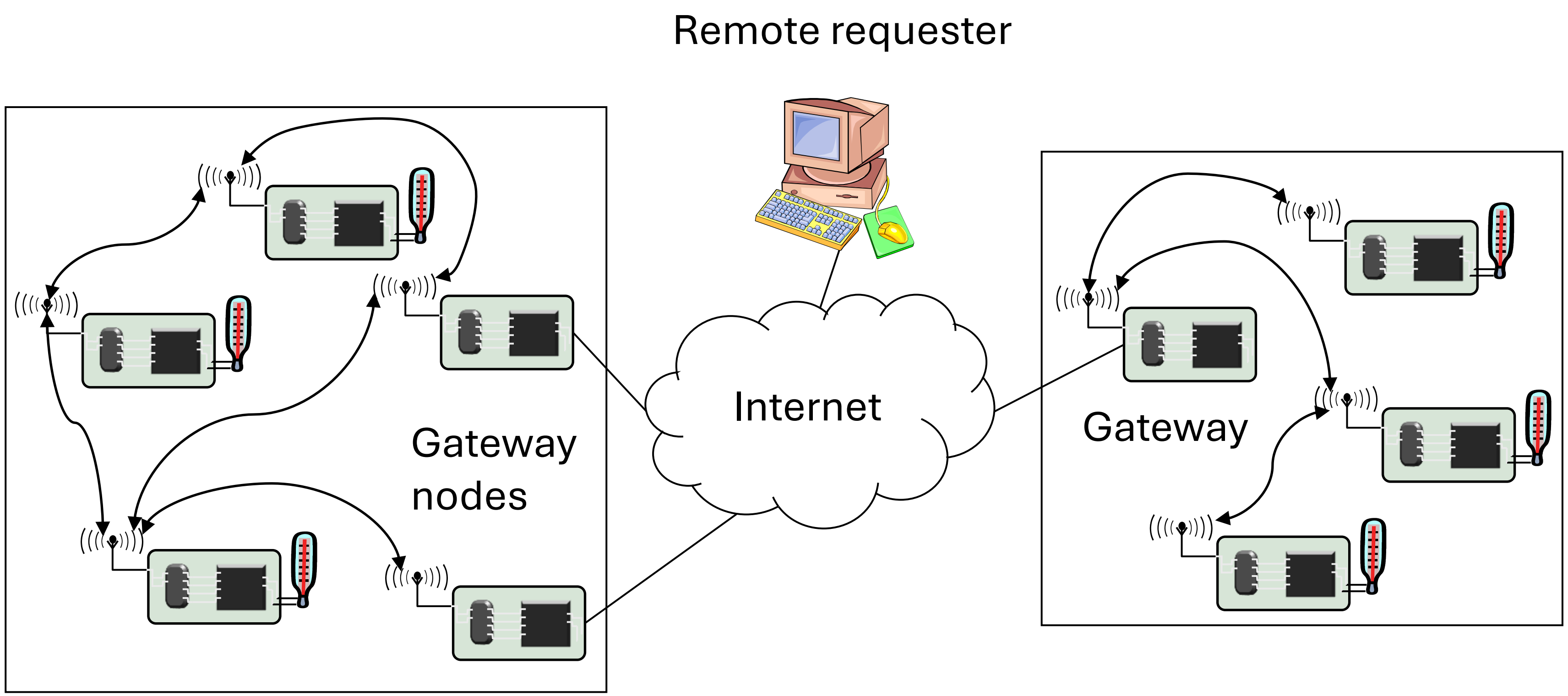
WSN to Internet communication

- Example: Deliver an alarm message to an Internet host
- Issues
  - Need to find a gateway (integrates routing & service discovery)
  - Choose “best” gateway if several are available
  - How to find Alice or Alice’s IP?



Internet to WSN communication

- How to find the right WSN to answer a need?
- How to translate from IP protocols to WSN protocols, semantics?



WSN tunneling

- Use the Internet to “tunnel” WSN packets between two remote WSNs